

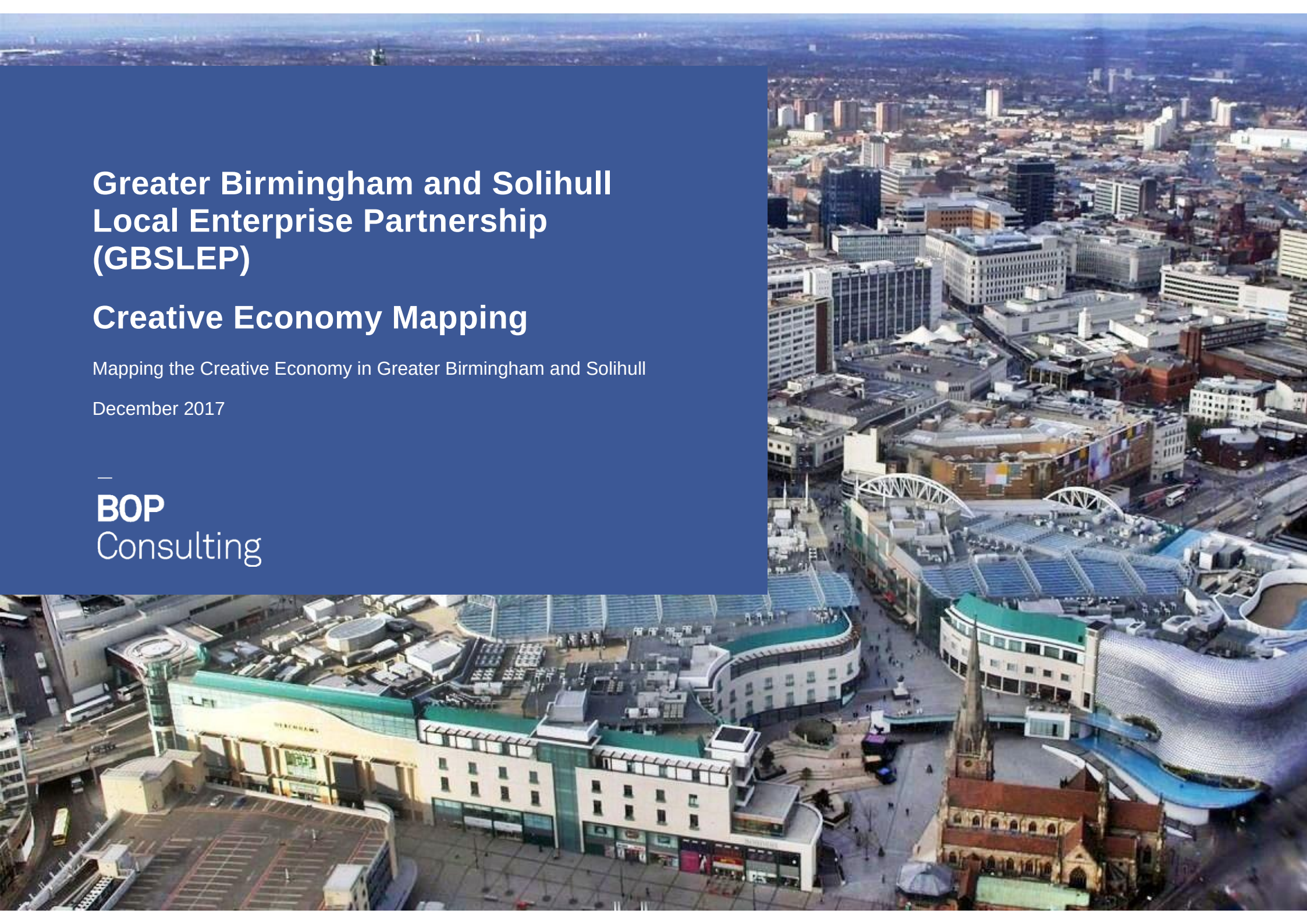
Greater Birmingham and Solihull Local Enterprise Partnership (GBSLEP)

Creative Economy Mapping

Mapping the Creative Economy in Greater Birmingham and Solihull

December 2017

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Contents

Acknowledgements	5
Executive Summary.....	6
1. Approach and methodology	20
1.1 GBSLEP geography	20
1.2 Methodology and terminology	21
1.3 BOP Background	23
2. Scale and Profile of GBSLEP Creative Industries	24
2.1 Number of creative enterprises in the GBSLEP area and comparable geographies.....	24
2.2 Profile of creative enterprises.....	26
3. Jobs and Skills in the Creative Industries	29
3.1 Number of Creative Industry Workers	29
3.2 Creative Economy Workers	31
3.3 Skill levels and requirements of Creative Industries workers.....	33
3.4 Vacancies.....	34
3.5 Diversity.....	36
3.6 Future jobs growth	36
3.7 Future skill needs.....	37
4. Value and Significance of Creative Economy.....	39
4.1 GVA of Creative Industries in Greater Birmingham	39

4.2 Creative employment in Greater Birmingham & Solihull.....	40
4.3 Significance of Creative Economy to GBSLEP Economy.....	41
4.4 Creative originals, content, services, experiences	42
4.5 Cluster analysis of Creative Industries in Greater Birmingham and Solihull	44
5. The competitive position of GBSLEP.....	49
6. Findings from Consultation	53
7. SWOT of GBSLEP Creative Economy	55
7.1 Strengths of GBSLEP Creative Economy	58
7.2 Weaknesses of GBSLEP Creative Economy	64
7.3 Threats to GBSLEP Creative Economy	67
7.4 Opportunities for GBSLEP Creative Economy	68
8. Analysis of findings.....	71
8.1 What are the main trends affecting the sector?.....	71
8.2 How are these manifesting themselves?.....	71
8.3 What is the impact of Brexit?	71
8.4 Which sub-sectors are most significant in GBSLEP and the West Midlands?	71
8.5 Which locations within the LEP area are significant (and showing fastest growth)?.....	72
8.6 Where are the main competitors (for investment)?	72
8.7 What are the key advantages of a West Midlands location for creative and digital businesses?	72

8.8 What are the key challenges to the sector in GBS LEP?.....	72
9. Conclusions	74
10. Recommendations.....	75
10.1 Productivity and skills.....	75
10.2 Infrastructure	79
10.3 Governance and culture of the cluster	80
Annex 1. Change in number of creative enterprises in GBSLEP local authorities, 2010-15	83
Annex 2 - Birmingham.....	88
Annex 3 - Bromsgrove	89
Annex 4 –Cannock Chase	90
Annex 5 –East Staffordshire	91
Annex 6 - Lichfield.....	92
Annex 7 - Redditch	93
Annex 8 - Solihull	94
Annex 9 - Tamworth	95
Annex 10 - Wyre Forest.....	96
Annex 11 – DCMS Creative Industries SIC Codes.....	97
Annex 12 - DCMS Standard Occupation Codes (SOC).....	98
Annex 13 – Tech Nation SIC codes	99

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BOP Consulting

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Executive Summary

INTRODUCTION

Greater Birmingham and Solihull Local Enterprise Partnership (GBSLEP) commissioned BOP Consulting to map the scale and output of the Creative Economy across the LEP area, comprising: Birmingham, Bromsgrove, Cannock Chase, East Staffordshire, Lichfield, Solihull, Tamworth and Wyre Forest.

This includes not just cultural organisations, creative and digital businesses and self-employed creative workers; but also, the thousands of people deploying their creative talents in other sectors of Greater Birmingham's economy.

We were excited to receive this commission. We believe it is a significant innovation in terms of studies of creative sectors in UK core cities, and one that provides policy makers, employers and investors with a clearer indication of the true scale and economic contribution of creative skills across the economy.

Our findings, summarised below, bear out the extent to which the LEP was right to take this wider view. Not only has our research and consultation shown the full scale and dynamism of a Cultural and Creative Industries sector which is not always as celebrated as that of other cities; but it demonstrates the level to which creative and digital innovation is embedded in other sectors, enhancing the potential growth of the whole Greater Birmingham economy.

Key findings

- The Creative Economy of Greater Birmingham generates £4.1 billion in GVA – 9% of total output of GBSLEP
- There are 50,000 Creative Economy workers in Greater Birmingham
- 5.6% of total workforce of GBSLEP is employed in the Creative Economy – a larger share of employment than Advanced Manufacturing & Engineering
- Number of Digital & Creative businesses increased from 5,100 to 6,150 between 2010 and 2015
- 18,400 people employed in Creative Services, generating £1.9 billion in GVA
- 16,850 people employed in Creative Content Production sector generating £1.4 billion in GVA
- A nationally significant cluster of Crafts businesses – over 11 times as concentrated in the centre of Birmingham as the national average
- Some of the UK's leading cultural organisations
- A talent pool of 32,000 people across Advertising & Marketing and IT, Software & Computer Services occupations, embedding innovation in all sectors of the GBSLEP economy

GROWTH OF THE CREATIVE INDUSTRIES

We have mapped the creative industries of the GBSLEP area across several dimensions:

- Scale and Profile of the Creative Industries: we have mapped the number and size of Creative Industries enterprises in Greater Birmingham, and how these numbers have evolved since 2010.
- Value and Significance of the Creative Industries: we have assessed the economic contribution of Creative Industries to the economy in Greater Birmingham, including:
 - The number of jobs within the Creative Industries in the GBSLEP area, broken down by sector and by local authority.
 - Changes in employment between 2010 and 2015.

- The relative importance of self-employment and freelance working across cultural organisations and the commercial creative sector.
- An estimate of Gross Value Added (GVA) output of Greater Birmingham's creative and cultural industries.
- Estimates of the number of vacancies within these industries and their skill requirements.

The Creative Industries¹ represent a significant proportion of the GBSLEP economy. Creative Industries businesses employed 27,500 people in 2015, representing 3.1% of total employment in Greater Birmingham, and generated £2.1 billion in GVA.

Between 2010 and 2015²:

- Number of businesses in the Creative Industries grew from 4,000 to 5,100
- Employment in the Creative Industries grew by 1,300 jobs
- Employment within IT, software & computer services grew by 1,200 jobs
- Employment within Design (including product, graphic and fashion design) increased by 1,100, from 450 to 1,550 jobs

¹ Our definition of the creative industries reflects that used by Department for Digital, Culture, Media and Sport (DCMS) in preparation of national economic estimates. This means that it includes the economic output of artists and cultural organisations, along with commercial enterprises in creative sectors ranging from advertising to television, and those sectors of the digital economy most reliant on creative inputs, including the production and distribution of video games and production of other kinds of entertainment software, and consultancy.

Therefore, we use the term 'creative industries' throughout this report to include artists and the cultural sector, and freelance workers and the self-employed as well as those in employment.

² All company and job numbers are rounded to the nearest 50

Some creative sectors have experienced a fall in employment between 2010 and 2015 – these include:

- Film, TV, Video, Radio & Photography (down 42 %, from 3,700 to 2,100 jobs)
- Publishing (down 28 %, from 1,200 to 900 jobs)
- Museums, Galleries & Libraries (down 12 %, from 1,700 to 1,500 jobs)
- Music, Visual & Performing Arts (down 5 %, from 2,100 to 2,000 jobs)

There is also a suggestion from consultation that increasing pressure on affordable studio space for artists and musicians may result in some displacement of those creative professionals – reflected in the fact that falls in employment in Birmingham city centre are countered by growth in some of the geography's towns and local authorities.

Therefore, falls in employment over the period 2010-15 do not necessarily indicate long term decline; and in some cases (particularly film and television), they may in fact be the precursor of new investment opportunities.

In drawing these conclusions, we would point out that these findings are the result of analysis of national statistics, not of a detailed census of companies in the sectors concerned. The mobile and contract-based nature of employment in some film, television and other creative arts may also contribute to the variance in numbers year on year.

But there are concerns about the scale of businesses within the Creative Industries in the GBSLEP area:

- The only sector in which more than 1% of businesses employed more than 50 people was Crafts – which, as defined by DCMS, can be interpreted as the manufacture of jewellery.

- Only 1% of businesses in Film, Television, Video, Radio & Photography employ 10 or more people, and no independent production company employs more than 50.
- The average size of creative businesses fell between 2010 and 2015, from 6.4 to 5.4 employees, against a UK sector average of over 7 employees. This may be a sign of economic dynamism in the sector, as established professionals set up new firms to take advantage of new opportunities. It would explain, in part, why growth in jobs has been slower than growth in the number of businesses. But there is also a risk that the small size of creative firms in Greater Birmingham reflects potential instability and lack of resilience to further economic shocks within the sector.

How should we interpret these results?

Some creative sectors in the GBSLEP area showed falls in employment and number of enterprises between 2010-2015 (see figure opposite.)

There are plausible reasons for these falls in employment:

- The **Audio-Visual sector** has suffered from a lack of commissioning infrastructure (and, therefore, spend) resulting from decisions by both BBC and ITV to centralise capacity in London. There are, however, signs that this process is beginning to be reversed – as outlined in the ‘Rebirth of a sector’ case study, below.
- **Publishing** has been disrupted by digital technology to a greater extent than other markets for creative content, and this is reflected in falls in company numbers across many core cities.
- Reductions in public sector expenditure have hit **Museums, Galleries & Libraries** particularly hard: the direct impact on employment is reflected in a ‘downstream’ reduction in demand for other creative services.

Creative freelancers and self-employed practitioners

We attempted to address concerns that employment data for the Creative Industries systematically under-reports the scale and contribution of freelance and self-employed workers in those sectors – particularly those involved in ‘core’ artistic practice, crafts and other sectors known to rely heavily on short-term contracts, such as film and television. To do this, we compared official labour market statistics from the Office of National Statistics (NOMIS) with alternative data. We constructed a database of over 69,400 LinkedIn profiles³ that referred to the geographies that compose the GBSLEP area. We included both profiles labelled as ‘freelance’ and as ‘self-employed’.

NOMIS showed a decline in the proportion of sole proprietors in Creative Industries, from 15 % of the creative workforce to 8 %, The LinkedIn data typically showed a higher proportion of freelance or self-employed workers in creative occupations and seems to more closely reflect observed patterns of employment in these sectors.

Unfortunately, it is not possible to bring together the NOMIS data and this alternative data in a way that provides a solid quantification of the number of creative freelancers in Greater Birmingham & Solihull. This is because of the high risk of double counting when attempting such a consolidation. Nonetheless, the alternative datasets do indicate that freelancing is a significant activity within the Creative Industries of Greater Birmingham & Solihull.

It is clear that freelancing is particularly important in some sub-sectors of the Creative Industries but features, at least to some extent, across the whole sector. When we combine those that report themselves as ‘freelance’ and ‘self-employed’, we find that these forms of employment

³ Recorded on 4 July 2017

are particularly important in Publishing; Film, TV, Video, Radio & Photography; Music, Visual & Performing Arts; and Crafts.

Birmingham's strong core of arts and cultural organisations

Birmingham has arguably the strongest cluster of major arts organisations in any English city outside London. This includes the City of Birmingham Symphony Orchestra, Birmingham Repertory Theatre, Birmingham Royal Ballet and the Ikon Gallery. The cultural offer drives the visitor economy, enables knowledge transfer within and between creative sectors and has the potential to further catalyse efforts to develop greater diversity within the cultural leadership and creative workforce of the city and the wider West Midlands region.

Arts and culture (live performance, museums, galleries and libraries) are important, both in transactional, relational and reputational terms. They provide a bridge between community activity, the outstanding cultural offer and commerce. Moreover, they are an important representation of the youth and diversity of the local population. Birmingham is both the youngest and most diverse of any city in Europe, with 40% of its population aged 25 or under.

There is evidence of spillovers between cultural organisations and the commercial creative sector – including innovative and international business models of arts organisations such as Ikon Gallery, with its large international specialist publishing business, and Stan's Café, whose innovative and scalable theatre show, "Of All The People In All The World" has been licensed as a format to other producers in New York, Melbourne, Buenos Aires and across Europe since its premiere in 2003.

Informal intermediaries

Leadership is not confined to large cultural organisations, nor to publicly funded ones. Birmingham Hippodrome is a commercial operation that plays a major role in offering greater access to its building and promoting diversity in cultural leadership and entrepreneurship amongst its neighbouring communities. Smaller festivals such as Flatpack and

independent organisations such as Ikon and Stan's Café act as informal intermediaries, providing mentoring and market information to less established companies and individual artists. They offer examples of arts organisations successfully diversifying their business model to reduce dependency on public subsidy, using a mix of subsidised and commercial activity to identify and develop new talent and help identify funding to secure and exploit IP created in the process. This expertise and knowledge is being transferred directly and indirectly to other creative practitioners and firms, through festivals, training programmes and workshops.

Clustering in the Creative Industries

Analysis of the concentration of businesses within and between creative sectors and local authorities within the GBSLEP area helps us to understand key strengths and the added value of the creative economy. We have used both Location Quotient (LQ) analysis of NOMIS and evidence from alternative data sources (eg, LinkedIn and Meetup) to establish the scale and location of creative clusters across Greater Birmingham.

Location quotient (LQ) is a way of quantifying how concentrated a particular industry, occupation, or demographic group is in a local geography, as compared to the nation. It is the standard technique for measuring the strength of an economic cluster. In this case, we have assessed employment by creative sectors in Greater Birmingham against national averages. LQ scores of 1.0 or above are indicative of clustering.

Cluster analysis reveals that there are concentrations of most of the Creative Industries groups across the Greater Birmingham & Solihull LEP are

Greater Birmingham has creative clusters in:

- **Crafts**, with an LQ of 6.2, is more strongly clustered in the GBSLEP area (and particularly in the city of Birmingham) than in any comparable LEP area;
- **Design** (comprising Product, Graphic and Fashion Design), which is showing extraordinary rates of growth in both the number of enterprises and employment in all local authorities in the GBS LEP area (averaging 238 % between 2010 and 2015);
- **IT, Software & Computer Services** - while this sector only just clears the threshold of 1.0 associated with clustering when analysed using the DCMS definition, the case that there is a digital cluster locally is strengthened by repeating the analysis on the Tech Nation sectors and deriving a score of 1.5.

We repeated the LQ analysis at the level of the local authorities that comprise GBSLEP. For the most part, we find that clustering is reasonably well dispersed across Greater Birmingham, with the same sectors that appear at the LEP area level associated with particular local authorities; but we also note:

- The economic contribution of creative businesses and workers is widely dispersed across Greater Birmingham & Solihull.
- Creative jobs and businesses are distributed across the whole area.
- The number of creative enterprises increased in every local authority within the GBSLEP area between 2010 and 2015.
- We found concentrations of different groupings of creative enterprises and workers within the local economies of most boroughs. There are different strengths within the creative industries across most areas of the conurbation.
- The Design and IT clusters are more widely geographically dispersed than the Crafts cluster, which is very concentrated upon Birmingham itself - and most likely, attributable to the Jewellery Quarter.

- We do see some different kinds of clusters at this more local level, e.g., Architecture in Lichfield.

Opportunity to drive inclusive growth

This distribution of sector specialisms and nascent clusters across the boroughs that make up Greater Birmingham & Solihull LEP area illustrate how some industries are increasingly looking outside the city centre for lower cost premises; but the continuing high rate of business and employment growth in the urban core suggests that this is part of a picture of more balanced growth, and not just displacement of arts and cultural businesses. As such, it represents a major opportunity for further, inclusive growth across the GBSLEP area.

Value and Significance of the Creative Economy

The Creative Economy represents a larger share of employment in the GBSLEP than either Advanced Manufacturing & Engineering or Transport & Logistics.

The economic impact of creative practice is not limited to either the cultural sector or the Creative Industries. Both within and beyond the Creative Industries, DCMS identifies Standard Occupation Codes (SOCs) associated with particularly creative practice to make an important distinction between the Creative Industries and the Creative Economy:

- The Creative Industries are composed of industries within a set of standard industrial classifications (SIC codes), in which the proportion of employees in creative practices passes a 30% threshold.
- Within these industries, there are both workers in the creative occupations and other so-called non-creative occupations.
- The Creative Economy includes the contribution of those who are in creative occupations (identified by SOCs) outside the Creative Industries, as well as within the Creative Industries.

We have analysed data on creative occupations held by the ONS Virtual Microdata Laboratory (VML). This allowed us to calculate the number of people in creative occupations outside the Creative Industries within the GBSLEP area. This is used as the basis of an analysis of the importance of the Creative Economy to Greater Birmingham and the wider West Midlands region, using national comparators. It also offers additional insight into the extent to which creative practice, and digital as an enabling technology within that, is a driver of innovation and growth across the whole economy of a city or city region.

The Creative Economy is a measure both of employment within the creative industries sector and the number of people in creative occupations in other sectors. This gives us a more accurate picture of the scale of the creative and digital talent pool across the GBSLEP area; and indicates the degree to which creative innovation is embedded within the economy as a whole.

Greater Birmingham has a large, growing and dynamic Creative Economy, employing nearly 50,000 people and generating £4.1 billion GVA per annum, 9% of the total GVA of £44.5bn. It employs 5.6 % of the total workforce in the Local Economic Partnership Area.

Why does this matter?

Creative Services (such as Advertising & Marketing) and IT, Software & Computer Services have in common a core of high level specialist digital skills, from data analysis to visualisation. Dispersal of these skills across industry sectors is associated with higher levels of innovation and productivity, and is therefore critical to the competitiveness and growth, both of the Creative Industries and the economy as a whole.

The balance of these occupations between the Creative Industries and the wider economy differed more than we expected. 80% of roles in Advertising & Marketing were found outside the Creative Industries, reflecting the fact that many small businesses and nearly all medium and large companies in all sectors will retain some marketing capacity 'in house.' However, 80% of employment in IT, Software & Computer

Services in Greater Birmingham & Solihull were within the Creative Industries, reflecting higher degrees of 'outsourcing' of technical, web design and analytical tasks to specialist agencies.

Design and crafts

The creative occupation with the next largest impact outside the Creative Industries was Design, with 2,000 employees in other sectors generating £138 million in GVA – 57 % of total Design output.

Perhaps more surprising is the impact of Crafts in the Creative Economy – 57 % of total output and employment, or £56 million per annum, and 1,150 jobs. This both reflects the narrow definition of Crafts applied by DCMS, and emphasises the relative strength of the Crafts cluster within Greater Birmingham's economy – explored in more detail below.

Creative supply chains

We have looked at our data results in another way to bring new insight to the strengths of the Creative Economy and Industries in Greater Birmingham & Solihull.

We have subdivided the Creative Industries into groups reflecting their outputs and value chains, rather than their skills inputs:

- Creative Content – the audiovisual and publishing sectors, primarily concerned with production and distribution of entertainment content in either physical, broadcast or electronic form
- Creative Experiences – live events, education, museums, galleries and libraries
- Creative Originals – design and production of artefacts, including jewellery and fashion
- Creative Services – advertising, marketing, architecture, graphic design, translation services and other publishing.

This analysis emphasises the scale, significance and value of the Creative Content production sector: despite recent falls in employment, it still accounts for more jobs (16,850) and output (£1.4 billion) than any other of the creative groupings.

It further emphasises the significance of Creative Services in the wider economy, with an output of £1.9 billion GVA per annum.

Digital Industries

This includes all the sectors of IT, Software & Computer Services within the DCMS definition of the Creative Industries and adds other elements of manufacturing and repair of computer equipment, telecoms, network and facilities management, data processing, web hosting and operation of web portals.

We have analysed the scale, profile and value of the Digital Industries to provide an insight into the dispersal of those industries across GBSLEP, and how that relates to concentrations of other creative industries. We have also weighted them as part of a merged definition of 'Digital & Creative Industries', to give a more complete sense of the value of the whole supply chain of web-based creative content and professional services within the GBSLEP economy.

The picture for Digital Industries was mixed. The number of digital businesses (as defined in the Tech Nation report) has grown by 27 per cent to 3,750; and their output, at £2.3 billion in GVA, was higher than that of the Creative Industries.

However, employment within the Digital Industries was down, by 3 per cent, to 22,350 – which reflects a steeper fall in average company size, from 7.8 to 5.9 employees, than that observed in the creative sector.

We don't have enough evidence to establish the cause of the decline – but observe that it reflects a trend across both West and East Midlands, and in other comparator LEAs such as West of England, but contrasts with strong growth in digital employment in Greater Manchester (51 %) and Cambridge and Greater Peterborough (36 %).

Regional clusters

There are three sectors of the Creative Industries which demonstrate scale, growth and clustering not only within the GBSLEP area but across the West Midlands:

IT, software and computer services

Within the DCMS definition of IT, Software & Computer Services, there were 14,000 workers in Greater Birmingham in 2015, generating 56 % of Creative Industries GVA - £1.2 billion in 2015.

This is an important sector across the West Midlands, with a further 8,450 employees in IT, Software & Computer Service in Coventry and Warwickshire and 2,600 in the Black Country contributing to a digital talent pool of more than 25,000 people within an hour's travel time.

From computer games to digital manufacturing

One key sub-sector within this grouping is the development and publishing of computer games. Analysis carried out on behalf of Coventry & Warwickshire LEP and UK Interactive Entertainment (UKIE) identified a national and internationally significant cluster of around 130 businesses, employing 3,500 people and generating £188 million in GVA, centred on Leamington Spa.

Within GBSLEP, games form an important and growing component of the high-end production sector for original digital content. And across the West Midlands, other industries, such as the Advanced Manufacturing & Engineering sector, are making increasing use of augmented and virtual reality (AR and VR) tools, often born out of games development skills and processes, to accelerate the growth of high value digital manufacturing.

Creative Services

We have already commented on the significance and scale of Advertising & Marketing within Greater Birmingham's Creative Economy. Employment within specialist Advertising, Marketing and PR agencies that form part of the Creative Industries in the GBSLEP area rose by a third – 700 jobs – between 2010 and 2015. The number of Advertising & Marketing workers within the Creative Industries also increased by 93% in the Black Country, and 114% in Coventry & Warwickshire over the same period.

Design

Employment in Design has increased by 238%, to over 1,500 jobs, between 2010 and 2015 in Greater Birmingham. And we have observed that 57% of design jobs and output in the GBSLEP area occur outside the Creative Industries. The number of designers in the Creative Industries has also increased in the Black Country LEP (245 % increase between 2010 and 2015) and Coventry & Warwickshire LEP (239% between 2010 and 2015).

Production of creative content

There are 22,000 jobs in creative content production across Greater Birmingham, generating £1.859 billion in GVA. Production of television, games and other creative content is concentrated in Birmingham's city core and in Solihull. When added to the games sector around Leamington Spa, it forms part of an emerging cluster of high-end content production that is significant at a national level.

Independent production in the West Midlands – rebirth of a sector

The closure of BBC Pebble Mill in 2004, following the closure of Central TV's Broad Street studio in 1997, heralded a difficult decade for broadcast production in the West Midlands. The loss of BBC's English Regions Drama hub, and an iconic location for daytime and news programming, and the dwindling of ITV Central to a single news studio, seemed to symbolise a loss of confidence in the region's TV production sector.

The decline began to be arrested by local indie production company, Maverick TV. As well as originating well known shows such as Embarrassing Bodies, Maverick produces BAFTA and Emmy award-winning interactive formats, chart-topping apps and games, innovative branded content and some of the UK's biggest YouTube channels, for broadcasters, agencies and brands including Channel 4, ASOS, Sky, AMVBBDO, Syco, Film4 and Microsoft.

In the last few years, a raft of returning industry veterans and new arrivals have bolstered the sector. Alex Fraser and Liza Abbott left Maverick in 2014 to establish 7 Wonder, a producer of non-scripted TV including new Channel 4 medical series, The Untreatables.

Another former Maverick TV executive producer, Colette Foster, whose track record includes 10 Years Younger, How to Look Good Naked, Embarrassing Bodies, Supersize versus Superskinny, The Sex Education Show, Gok's Fashion Fix, Secret Eaters, Mary's Bottom Line, Superscrimpers, Katie Piper's Bodyshockers, and BBC 2's The House That £100K Built and Restoration Home. Brummie born-and-bred, she returned from London to set up Full Fat TV in September 2015. Its website proudly proclaims that it is, 'A factual entertainment TV company based in Birmingham.' Full Fat produces award winning shows including Joe Hicks: The Body Coach for Channel 4 and Tony Robinson's Coast to Coast for Channel 5.

To this we can add the efforts of BBC's much-expanded (and still growing) Midlands operation, centred on the Mailbox in Birmingham. The long-running daytime serial drama, Doctors, has given BBC Studios experience of producing many hours of drama at low cost – which it is able to pitch to other UK and international broadcasters. Father Brown, also produced in Birmingham, now sells to an extraordinary 220 territories. The arrival of BBC3 in 2018 will be a magnet for young content makers across a variety of genres, including comedy and factual programming.

The independent production sector itself is aware that it has turned a corner; and the wider digital and creative industries in the West Midlands are keen to seize the wider opportunities for commissioning of new forms of content and technical services that a relocation of Channel 4 would present.

Comparison of the GBSLEP Creative Industries offer with other core cities and knowledge clusters

We offer a comparison with the performance of other comparable and competitive LEP areas in supporting this high growth, high value sector of the knowledge economy.

We use a combination of measures of the scale, significance, output and clustering to compare and contrast the performance of the Creative Industries in Greater Birmingham with other urban centres in the UK.

We compared the GBSLEP area with a basket of comparator cities, composed of English core cities outside of London with established creative sectors, including Manchester, Bristol and Leeds, and city regions with strong innovation infrastructure, such as Oxford and Cambridge. We find that while Greater Birmingham outperforms some other geographies on a range of indicators, performance elsewhere exceeds that of Greater Birmingham, suggesting some degree of unfulfilled potential for GBSLEP to address.

- The Greater Birmingham & Solihull area outperforms a number of LEPs but lags behind Oxfordshire, West of England (including Bristol), and Greater Cambridge and Greater Peterborough in the proportion of local enterprises that operate within the Creative Industries.
- Although the proportion of creative enterprises within Greater Birmingham's economy is on a par with many competitors, including Greater Manchester and Leeds City Region, if it were to match the performance of Oxfordshire, it would add 3,965 enterprises to the business base of the GBSLEP area.

- If levels of job creation in Greater Birmingham's Creative Industries were to match the performance of the sector in the strongest LEP (Oxfordshire), just over 30,000 jobs would be added to the economy in the GBSLEP area.⁴

SKILLS

Low levels of qualifications may be limiting growth

The proportion of the population of the West Midlands with qualifications at level 4 and above correlates with our finding that qualification levels are typically lower in cultural and creative businesses in the West Midlands than elsewhere in the UK. At 36 % of cultural and creative businesses across the UK, more than 80 % of staff hold Level 4 qualifications. In contrast, in only 25 % of such businesses in the West Midlands is this the case.

This would suggest that there are fewer highly skilled individuals in the jobs market in Greater Birmingham than in some comparator cities.

Of course, it is the case that many people make fantastic contributions to the Creative Industries without holding Level 4 qualifications, but we also take higher level qualifications as a proxy for skill and adaptability. With Greater Birmingham also showing lower concentrations of creative businesses than competitor LEPs – reflecting a less strongly 'clustered' sector, with less potential for exchange of knowledge and skills to promote innovation – this carries the risk that employers in the region will continue to struggle to identify and recruit people with relevant skills and experience for the new jobs that will emerge over the next two decades.

⁴ This is the total number of workers in Greater Birmingham and Solihull multiplied by the proportion of creative industry workers in Oxfordshire minus the number of such workers already in Greater Birmingham and Solihull.

The final risk attached to lower levels of creative skills – particularly when combined with observed lack of scale within creative businesses in Greater Birmingham– is an increased threat from automation of more ‘process driven’ jobs within the sector.

Current vacancies in the Creative Economy

We have analysed LinkedIn to better understand the volume and context of current vacancies within three sectors of particular importance to the Greater Birmingham & Solihull area: Advertising & Marketing, Design, and IT, Software & Computer Services.

- Of the three sectors analysed, the largest number of vacancies were in **IT, Software & Computer Services** - which is consistent with its position as the largest creative sector in Greater Birmingham, both in terms of number of workers and enterprises.
- **Design** vacancies speak to the relevance of this discipline across a wide range of settings - engineering, mechanical, architectural, and web.
- **Marketing** seemed to be generating many more vacancies than Digital Marketing or Public Relations, reflecting the high levels of demand for people with general marketing skills in companies outside the creative industries (as opposed to the more specialist roles that exist mostly within Advertising & Marketing agencies).

Despite the low skills levels, we were unable to find evidence that creative businesses within Greater Birmingham encounter greater problems in recruitment than their peers in competitor cities.

Future jobs growth/future skills needs

We would single out three areas in which trends analysis suggests there will be continuing growth in jobs:

- Data analysis and behavioural insights, already identified as a strength within the region, will become increasingly important in

the context of rising adoption of artificial intelligence and machine learning

- IT, software development and computer services
- Multitalented and multiplatform content producers

These ‘horizontal’ capabilities will be increasingly important in the production of creative media. The availability of people with a mixture of data and creative skills has been critical in developing expertise in ‘programmatic’ advertising, using algorithms to automate the buying, placement, and optimisation of media inventory that works in real time.

Multiplatform skills are essential to the development of newer immersive forms, both for entertainment and other industrial and social applications. As market demand for augmented and virtual reality grows, producers will increasingly depend on ‘hybrid’ skills of analysis, insight and software development.

Trends analysis

The Creative Economy in Greater Birmingham is relatively well positioned to take advantage of some trends affecting the Creative Industries. The focus that local universities have given to data analysis, behavioural insights, human computer interaction and machine learning, if transferred through graduate recruitment and collaborative research, should allow businesses across all sectors to address:

- Digital disruption of production: e.g., data analysis and artificial intelligence (AI) dramatically changing patterns of work in advertising; and digital manufacturing (‘fab labs’) transforming design, prototyping and production processes. The largest effect will be felt in sectors outside the Digital & Creative Industries: for example, agencies and individuals with those skills are already an indispensable part of the supply chain for Advanced Manufacturing & Engineering.
- Digital disruption of consumption: consumers of entertainment content are moving relentlessly online, with fragmentation of

audiences across platforms. Although often accompanied by growth in overall audiences, online revenues do not yet replace those in 'traditional' media – sectors such as music, journalism and publishing are having to find new business models to adjust to those changes.

- Much greater interest in 'immersive tech' from government and private investors. Immersive tech describes hardware and software used to create the perception that a digitally-created experience is 'real' - including augmented reality (AR), virtual reality (VR) and haptics (touch). It has applications both in entertainment and other high growth verticals (Advanced Manufacturing & Engineering, Transport & Logistics, Health & Life Sciences) and the Visitor Economy.

The existing sector mix in Greater Birmingham favours disciplines such as Advertising & Marketing, IT, Software & Computer Services, Design and specialisms in games and serious games that should all be able to adapt quickly to these new demands, and catalyse adoption of innovation in areas such as digital manufacturing.

Threats in an uncertain external environment

In an increasingly straitened investment climate, there is a risk that the city region's lack of a defined image as a creative hub, comparative lack of highly qualified people and scarcity of examples of businesses which have successfully 'scaled' will hinder attempts to attract new businesses to invest in the area.

And whilst investment in new infrastructure may bring jobs and opportunity, there is no strategy or policy in place to deal with disruption and displacement brought about by large scale plans for new infrastructure and commercial pressure on affordable workspace in existing Creative Industries clusters.

That includes the risk, highlighted by industry surveys in other core cities such as Greater Manchester, that HS2 may accelerate a 'brain drain' to

London, accentuating shortages of key skills and inflating salaries, that could offset any potential gains from inward investment.

The uncertainties of Brexit present some specific and painful risks to creative businesses, whether they export or not. Should Brexit result in restrictions on freedom of movement, it may further reduce access to talent, accentuate skills gaps and shortages, and inflate salaries as employers in Greater Birmingham and across the UK compete for a narrower pool of skilled employees.

Post-Brexit tariff and non-tariff barriers may limit growth of exports to Europe, and make overseas touring more expensive and difficult. Similarly, cultural organisations may find it harder to attract international talent, and to get the necessary visas for them to perform.

Compliance with multiple regulatory regimes post-Brexit - specifically, the EU's General Data Protection Regulation (GDPR) - and conflicting requirements for managing customers' data in a cross-border environment, particularly with US platforms, could increase costs and risks for IT, advertising and marketing businesses.

CONCLUSIONS

Our overall assessment is that the potential of the Cultural and Creative Industries sectors in Greater Birmingham remains unfulfilled. The rate of employment growth within these sectors between 2010 and 2015 was 8 % lower than in competitor cities. However, we see no reason why Greater Birmingham could not raise its rate of employment growth to match that achieved in other equivalent city regions.

Facing up to challenges

But GBSLEP and its partners in the West Midlands Combined Authority and local authorities still need to commit to shared actions to address some shortcomings.

- The diversity of Greater Birmingham's population is not fully reflected by or taken advantage of in its cultural and creative

industries. Cultural and creative employers of all sizes need help, first to acknowledge their shortcomings in this respect, and then to act on them.

- Uneven growth between sectors and across geographies means that the benefits of inclusive growth are not equally shared. And transport and communications within and across the LEP area are less coordinated than the national and trunk routes in and out. A better integrated strategy for transport, housing, culture and liveability could point up the 'quality of life' benefits for both existing residents and inward investors. Within that lies the opportunity to spread the benefits of inclusive growth to the other boroughs of GBSLEP, and to address the shortfall in residents with Level 4+ skills.
- The lack of large creative businesses, especially indigenous ones, presents barriers to the growth of microbusinesses, both in a constricted value chain, increased price competition, lack of role models for creative business leaders and fewer examples of successful entrepreneurs who have exited and are able and willing to invest in other new business propositions.

Some of these shortcomings can be addressed through inward investment, though not all: the city region's self-image is defined by a lack of self-confidence amongst creative businesses and cultural organisations.

Addressing weaknesses

To address this, steps need to be taken to address:

- Productivity and skills, building on strengths in key Creative Industries sectors, mitigating risks of further decline in weaker sectors and addressing the need to build social capital to promote inclusive growth. This calls for focused investment to increase the diversity of the Cultural and Creative Industries.

- Infrastructure - workspace, planning of new developments, transport and digital connectivity - including that linked to development of new cultural quarters across Greater Birmingham.
- Consultation revealed that cultural organisations and creative businesses across Greater Birmingham are less assertive about the benefits of their location, more dependent on the public sector for both funding and industry coordination and more critical of local infrastructure. Concerted action, coordinated between the public sector, informal intermediary organisations and creative businesses, is needed to address these concerns, to boost the confidence of the sector, increase its diversity and to convene new and more representative sector structures across the West Midlands.

Greater Birmingham – a great location

The GBSLEP area's central location offers a large hinterland, which in turn presents great opportunity to advertising and marketing businesses, reflected in the scale of their contribution to the creative economy.

Birmingham has a nationally significant cluster of major arts organisations – the strongest of any English core city outside London.

And it enjoys great transport connectivity – it is at the centre of national road and rail networks

Greater Birmingham offers much lower business premises costs than other competitor creative cities (eg London, Bristol, Oxford, Cambridge, Edinburgh, even Manchester and Leeds), making it cheaper to set up and run a business; whilst a mix and sufficient supply of different types of housing at affordable prices make it possible for young people to purchase a family home near to their place of work.

Capturing the momentum in high end content production

The high-end content production sector remains a strategic priority for Greater Birmingham and the West Midlands. LEPs across the region worked closely together in preparation of the response to consultation on the relocation of Channel 4. There is a shared understanding that this sector plays a key role in catalysing the aspiration of the region's 'young, diverse and digital' population; and that public service broadcasters, in particular, have a responsibility to offer better representation of the region and its citizens. This increased engagement has at once raised the profile of Greater Birmingham as a cultural catalyst, and a potential source of new audiences and new business models; and the awareness of politicians and people alike of the economic, social and cultural dividend that can result from investment in jobs and skills in its Creative Industries.

Turning 'good' into 'great'

In our view the situation around high end content production is a barometer for Greater Birmingham's Creative Economy. The Creative Content sector has reached an 'inflection point'. The official statistics end at a point where the independent television and film sector was just starting to witness a rebirth. The area has seen both a resurgence in confidence, new investors (BBC THREE), skills (BBC Academy), start-ups (7Wonder, Full Fat and many others) and plans for big capital investments that together have the potential to re-establish Birmingham and the surrounding city region as one of the major centres of content production outside London. It is now critical that policy should address how contraction can be reversed.

To that end, GBSLEP and other actors should combine to deliver a strategy for growth including some or all of: West Midlands Film Fund, Mercian Studios, expanded production capacity at the BBC and UK Central, relocation of Channel 4 and lobbying for higher and more tightly enforced quotas for 'out of London' production.

During the time that Greater Birmingham laboured under a relative lack of opportunity from 'mainstream' commissioning, businesses, universities and the public sector combined to address that challenge in new ways. This resulted in innovative and important new business models for the Creative Industries and the economy as a whole.

Lack of commissioning infrastructure forced creative content companies to work with a wider range of partners, and across digital platforms, in a way that represents the future of TV, rather than its past. Games companies forced to develop their own tools to create immersive experiences in augmented and virtual reality have found a market for those applications and processes in the wider digital manufacturing sectors.

And local advertising and marketing professionals have translated expertise in data analysis and behavioural insight into a business model better suited for programmatic advertising, which it will take an increasing share of revenue from advertising that sits alongside video content across a range of platforms.

In the same way, taking a regional view of the Creative and Digital Industries across the West Midlands can help turn some 'good' investment propositions into 'great' examples of smart specialisation - for example an internationally competitive games cluster and capacity within innovative content production across a range of platforms. Whether Channel 4 relocates to the region or not, Greater Birmingham's Creative Economy reflects the alignment of a high value and innovative production sector (from animation and games through to independent television production and film) with a 21st century, digital way of doing things. This, added to the advantage of a young, diverse talent pool capable of understanding the differing needs of international markets, should make location in Greater Birmingham an attractive option for forward-looking content businesses.

1. Approach and methodology

1.1 GBSLEP geography

1. The geographical focus of this cultural and creative mapping are the local authorities that compose GBSLEP, which are:
 - Birmingham
 - Bromsgrove
 - Cannock Chase
 - East Staffordshire
 - Lichfield
 - Solihull
 - Tamworth
 - Wyre Forest

Figure 1. Geography of Greater Birmingham and Solihull LEP



1.2 Methodology and terminology

1.2.1 Creative Industries

Our estimates of the scale of Cultural, Creative and Digital Industries in Greater Birmingham follow the sector definition used by the Department for Digital, Culture, Media & Sport (DCMS) to define ‘the Creative Industries’. The industries included are defined by Standard Industrial Classifications (SIC). The determination of which activities constitute the Creative Industries is determined by the proportion of creative workers within them, as defined by Standard Occupational Codes (SOCs) associated with creative practice.⁵

This report uses the term ‘Creative Industries’ throughout to describe the economic and employment contribution of cultural, creative and digital activities within the DCMS definition.

1.2.2 Standard Industrial Classifications (SIC) codes

2. The DCMS groups Standard Industrial Classification (SIC) codes to define the following Creative Industries sub-sectors:

- Advertising and Marketing
- Architecture
- Crafts
- Design: Product, Graphic and Fashion Design
- Film, TV, Video, Radio & Photography
- IT, Software & Computer Services
- Publishing

- Museums, Galleries & Libraries
- Music, Visual & Performing Arts

A full list of Creative Industries SIC codes can be found in the appendices.

1.2.3 Standard Occupation Codes

3. Both within and beyond the Creative Industries, DCMS identifies Standard Occupation Codes (SOCs) associated with particularly creative practice. A full list can be found in the appendices.

1.2.4 Creative Economy

DCMS makes an important distinction between the creative industries and the creative economy.

- The Creative Industries are composed of the SIC codes, as defined above. Within these industries, there are both workers in the creative occupations highlighted above and other so-called non-creative occupations.
- The Creative Economy includes the contribution of those who are in creative occupations (SOCs) outside the creative industries, as well as the creative industries.

We have included an analysis of the ‘Creative Economy’ of GBSLEP. To do this, we have analysed data on creative occupations held by the Office of National Statistics’ (ONS) Virtual Microdata Laboratory (VML). This analysis allowed us to calculate the number of people in creative occupations outside the Creative Industries within the economy of the Greater Birmingham.

⁵ The number of creative jobs in each industry is divided by the total number of jobs in that industry. Industries (SIC07) which have more than 6,000 jobs and a “creative intensity” of more than 30 per cent are considered as candidates for inclusion. For full detail on this DCMS methodology: https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/499683/CIEE_Methodology.pdf

1.2.5 Why do we include creative occupations outside the Creative Industries and 'non-creative' digital activity in our analysis of Cultural and Creative Industries in GBSLEP?

The Creative Industries are defined by high concentrations of creative workers but workers undertaking associated creative practices are not restricted to the Creative Industries. Therefore, to capture the full economic impact of these creative practices, we need to also take account of workers undertaking these practices outside of these industries. The combination of the output of these workers and the creative industries forms the Creative Economy.

While this combination of the Creative Economy and Creative Industries is the dominant method for understanding the economic impact of creativity upon the UK economy, there are some important creative industries that see themselves as being defined, in part, by SICs that extend beyond those SICs allocated to the creative industries by DCMS. In particular, Tech Nation have identified a broader grouping of SICs as defining 'Digital Industries', which we have analysed in the context of Greater Birmingham.

1.2.6 Literature review

We have undertaken a literature review, taking into account the activities of arts and cultural organisations and other work being done on cultural and creative tourism. We have also reviewed the latest policy recommendations for growing local creative clusters. This takes into account priorities within the Government's Industrial Strategy White Paper, recent statements from Sir Peter Bazalgette's independent review of creative clusters, and GBSLEP's possible response.

1.2.7 Findings from consultation

We have complemented the desk research with interviews and insights from three workshops with local practitioners and experts on the cultural and creative industries of Greater Birmingham and Solihull.

1.2.8 Competitor LEPs

To better understand the relative performance and significance of Greater Birmingham's Creative Industries, we have compared their performance with the average for the UK (see "GBS LEP vs other LEPs", below). Because London so dominates the UK's Creative Industries in both scale and output, and therefore skews any national comparisons, we have also analysed GBSLEP's performance against areas within each of three different LEP typologies:

(a) Other Local Enterprise Partnerships (LEPs) in the West Midlands:

- Black Country
- Coventry & Warwickshire

(b) LEPs that - like GBSLEP - cover the geography of an English core city.

During 2012 the first wave of 'city deals' recognised these eight cities as "the largest and most economically important English cities outside of London." In addition to Birmingham, they are:

- Derby, Derbyshire, Nottingham and Nottinghamshire LEP ("D2N2")
- Greater Manchester LEP
- Leeds City Region LEP
- Liverpool City Region LEP
- North East LEP
- Sheffield City Region LEP
- West of England LEP

(c) Other English LEPs that represent world-leading knowledge centres outside London:

- Cambridge & Greater Peterborough LEP
- Oxfordshire LEP

1.3 BOP Background

BOP is a specialist Creative Industries consultancy, established in 1997 and with offices in London, Birmingham, Edinburgh, Liverpool and Shanghai. We understand how culture and the Creative Industries enrich people's lives; how venues and commercial premises help shape places; and how together they make modern economies more competitive. We are expert in producing strategies to optimise the economic and social value and benefit of both.

BOP has played a critical role in defining the role of Creative Industries in economic development through strategies and toolkits for organisations including the United Nations Organization for Economic Co-operation and Development (OECD); the World Intellectual Property Organization (WIPO); the British Council; Arts Council England (ACE); and the Department for Digital, Creative Media & Sport (DCMS)'s Creative Industries Council.

We have also played a key role in defining the benefits of creative and cultural developments in economic terms. We introduced HM Treasury's 'Green Book' standard to this sector; and carried out the impact assessment of £1.4 billion of HLF-funded projects. Arts Council England asked us to prepare national guidance on how to commission economic impact studies; we have completed economic impact studies for over 70 venues and festivals.

BOP specialises in producing place-specific strategies for the development of Creative Industries clusters and capital projects. Our track record of advising local authorities and commercial developers which are planning to invest in capital developments focused on growth of the Creative Industries is second to none. BOP often works in a regeneration context, so we are used to articulating all the direct/indirect and measurable/qualitative ways in which the Creative Industries drive local economic development.

2. Scale and Profile of GBSLEP Creative Industries

This chapter looks at the number of creative industry enterprises within the local authorities that compose GBSLEP, the size of these businesses and how these numbers evolved between 2010 and 2015.

We go on to compare the evolution of the Creative Industries with that of the GBSLEP economy as a whole; and with comparable LEP areas.

2.1 Number of creative enterprises in the GBSLEP area and comparable geographies

Figure 2 (opposite) shows:

- On the DCMS definition, there were over 5,100 creative industries enterprises in Greater Birmingham and Solihull in 2015.
- This represented 8% of all enterprises in Greater Birmingham⁶ – which we compare against comparator LEPs later in this report.
- Over half of these enterprises are in the sector of IT, Software & Computer Services.
- The number of Creative Industries enterprises, according to the DCMS definition, increased by 26% between 2010 and 2015.
- When the extended Tech Nation definition of digital is applied, there were nearly 3800 enterprises in the digital sector in Greater Birmingham and Solihull - a 27% increase on 2010.

Figure 2. Creative Industries enterprises in the GBSLEP local authority areas in 2010 and 2015

All GBSLEP boroughs	2010	2015	% change ⁷
Advertising and Marketing	450	500	9
Architecture	250	300	42
Crafts	150	150	-18
Design: Product, Graphic and Fashion Design	350	450	28
Film, TV, Video, Radio & Photography	350	400	6
IT, Software & Computer Services	1,950	2,800	42
Publishing	150	200	13
Museums, Galleries & Libraries	0	0	50
Music, Visual & Performing Arts	400	400	-1
DCMS total	4,000	5,100	26

Source: NOMIS, ONS, BOP Analysis

⁶ This is based on a total business base of 66,120 enterprises.
<https://www.nomisweb.co.uk/reports/lmp/lep/1925185545/report.aspx>

⁷ Enterprise and employment data for 2010 and 2015 are rounded to nearest 50, while percentage change figures are calculated using the non-rounded figures.

2.1.1 Digital and Creative Industries

We have also analysed the contribution of a broader definition of the Digital Industries. This is taken from Tech Nation and is composed of the following four-digit SIC codes:

Figure 3. **Tech Nation SIC codes**

SIC code	Description	Included in DCMS Creative Industries definition?
2620	Manufacture of computers and peripheral equipment	N
5821	Publishing of computer games	Y
5829	Other software publishing	Y
6110	Wired telecommunications activities	N
6120	Wireless telecommunications activities	N
6130	Satellite telecommunications activities	N
6190	Other telecommunications activities	N
6201	Computer programming	Y

	activities	
6202	Computer consultancy activities	Y
6203	Computer facilities management activities	N
6209	Other IT & computer service activities	N
6311	Data processing, hosting & related activities	N
6312	Web portals	N
9511	Repair of computers & peripheral equipment	N

Where we include other 'digital' activity in our analysis, taken from the wider range of IT-related industries used in the Tech Nation report, we use the term 'Digital & Creative.'

Using this definition, there were 3,765 digital enterprises in Greater Birmingham and Solihull in 2015, up from 2,955 in 2010 – an increase of 27%, in line with the average for core cities across the UK. Combining DCMS and Tech Nation definitions and removing duplicates gives us a combined total of 6150 Digital & Creative Industries enterprises in GBSLEP. (rounded to the nearest 50).

2.2 Profile of creative enterprises

We now review the size of enterprises in the Cultural and Creative Industries in Greater Birmingham.

The table below shows:

- The significant majority of Cultural and Creative Industries enterprises in Greater Birmingham are microenterprises (businesses with fewer than ten employees). No creative sector has less than 64% of enterprises at this scale.
- Crafts is the only sector in which more than 1% of businesses has more than 50 employees. This is also the sector with the largest number of sole proprietors - suggesting that this sector has some divergence between a relatively large number of very small enterprises and a relatively large number of larger enterprises.
- The highest proportion of microenterprises comes in IT, Software & Computer Services, with 91% of businesses at this scale.

Figure 4. **Proportion of GBSLEP Businesses by Creative Sector of Different Enterprise Sizes**

	Sole proprietor	Micro (0 – 9 workers)	Small (10 - 49 workers)	Medium (50 - 249 workers)
Advertising & Marketing	8%	83%	9%	1%
Architecture	9%	84%	6%	1%
Crafts	22%	64%	11%	3%
Design: Product, Graphic and Fashion Design	12%	84%	4%	0%
Film, TV, Video, Radio & Photography	17%	82%	1%	0%
IT, Software & Computer Services	3%	91%	4%	1%
Publishing	13%	82%	5%	0%
Museums, Galleries & Libraries	0%	67%	33%	0%
Music, Visual & Performing Arts	21%	73%	5%	1%
DCMS Total	8%	86%	5%	1%
Tech Nation	4%	90%	5%	1%

Source: NOMIS, ONS, BOP Analysis

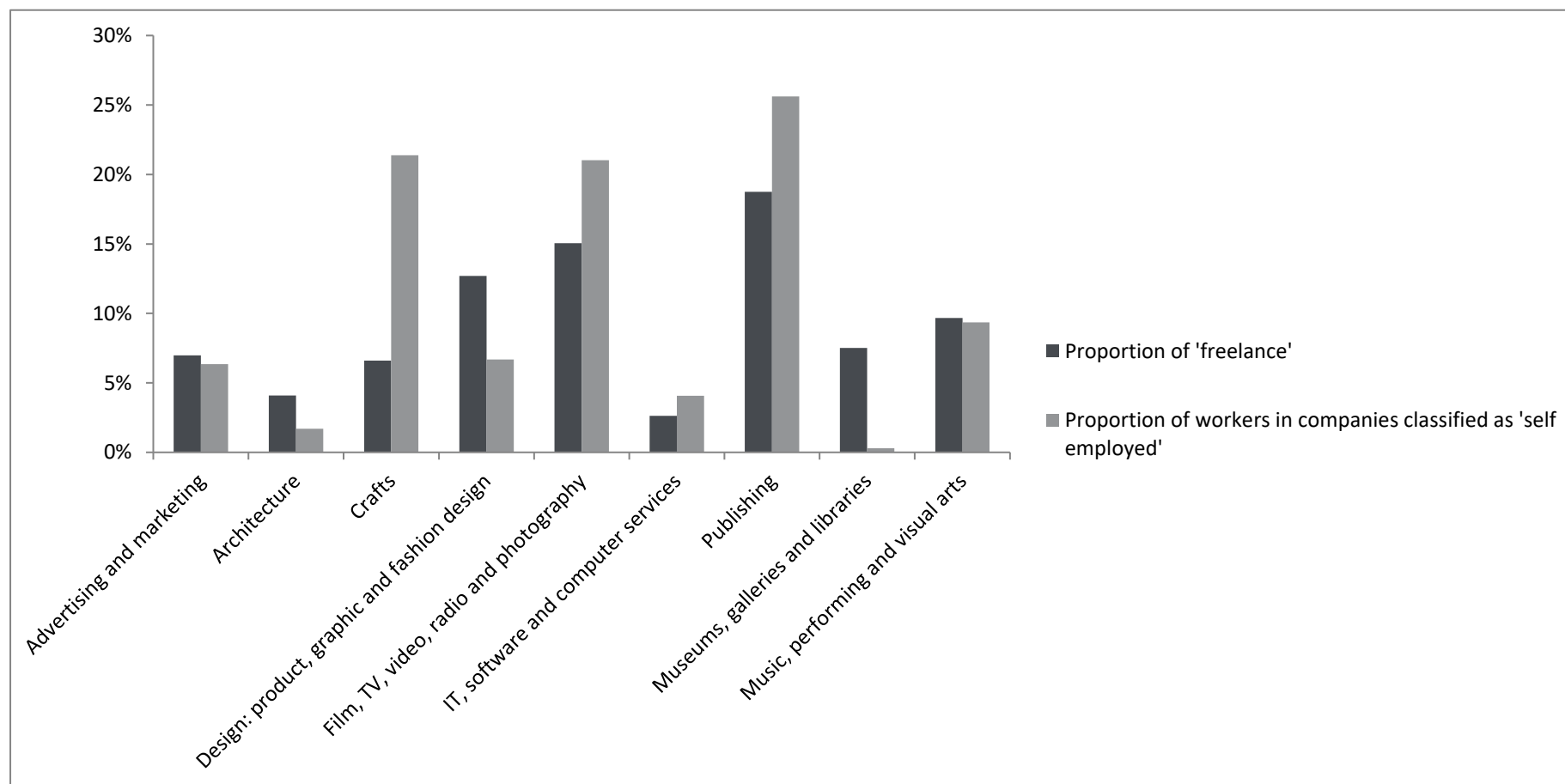
2.2.1 Freelancers and self-employed practitioners

- The proportion of sole proprietors, despite its recent decline from 15% of the creative workforce to about half that, gives us some sense of the importance of freelancers in the Creative Industries. But we explore alternative data, with a broader definition of freelancers, to better understand this importance.
- Unfortunately, it is not possible to bring together the NOMIS data and this alternative data in a way that provides a solid quantification of the number of creative freelancers in Greater Birmingham and Solihull. This is because of the high risk of double counting when attempting such a consolidation. Nonetheless, the alternative datasets do indicate that freelancing is a significant activity within the creative industries of Greater Birmingham and Solihull.
- This dataset is composed of LinkedIn profiles within the geographies that compose Greater Birmingham and Solihull. Our definition of freelancer for these purposes is: those LinkedIn profiles labelled as 'freelance' and those labelled as 'self-employed'.
- The results of this analysis are presented in Figure 5, below. It is clear that freelancing is particularly important in some parts of the creative industries but features, at least to some extent, across all

parts of the Creative Industries. When we combine the percentages that report themselves as 'freelance' and 'self-employed', we find that these forms of employment are particularly important in:

- There are 3,511 LinkedIn profiles in Greater Birmingham and Solihull in publishing sectors, of which 1,557 (44%) describe themselves as freelance or self-employed.
- There are 2,651 LinkedIn profiles in Greater Birmingham and Solihull in sectors associated with Film, TV, Video, Radio & Photography, of which 956 (36%) describe themselves as freelance or self-employed.
- There are 4,341 LinkedIn profiles in Greater Birmingham and Solihull associated with Crafts, of which 1,215 (28%) describe themselves as freelance or self-employed.
- These results have some consistency with the NOMIS data on sole proprietors, in which Crafts (22% of enterprises) and Film, TV, Video, Radio & Photography (17% of enterprises) feature among the three sectors with the highest proportions of sole proprietors.
- The third of these sectors is Music, Visual & Performing Arts (21% of enterprises), which also has a relatively large number of LinkedIn profiles labelled as freelance and/or self-employed (19%).

Figure 5. **Proportion of LinkedIn profiles in Greater Birmingham described as Freelance or Self-Employed by Creative Sector**



Source: NOMIS, ONS, BOP Analysis

3. Jobs and Skills in the Creative Industries

This chapter looks at the number of jobs within the creative industries in the GBSLEP area. We compare Greater Birmingham's Creative Industries with those of other core cities, and analyse how both the performance of the sector has changed in absolute and relative terms between 2010 and 2015. We also look at the number of vacancies within these industries and their skill requirements.

3.1 Number of Creative Industry Workers

The number of workers within the Cultural and Creative Industries in Greater Birmingham and Solihull increased by 5% between 2010 and 2015, when creative industries employment represented 3.1% of all jobs.

Much the largest sector of employment in 2015 was IT, Software and Computer Services with 14,100 workers - a 10% increase on 2010. When we look at the broader definition of the digital sector used by Tech Nation, this employment footprint expands to 22,400.

Design was a sector of particularly rapid growth over this period – adding a net 1,000 jobs to increase employment to over 1,500 – a 238% increase. Advertising (an additional 700 jobs, 33% growth) and Architecture (an increase of 470 workers, or 41% of total employment in that sub-sector) were the next most rapid sectors of expansion.

However, some sectors showed a decline in employment over this period:

- Film, TV, Video, Radio & Photography (down 42%)
- Publishing (down 28%)
- Museums, Galleries and Libraries (down 12%)
- Music, Visual and Performing Arts (down 5%)

While still accounting for a large number of jobs (22,400), the Tech Nation sectors also saw a 3% fall in employment.

Figure 6. **Change in employment in Creative Industries sectors in Greater Birmingham in 2010 and 2015**

Creative Industries employment in GBSLEP	2010	2015	% change 2010 - 2015
Advertising & Marketing	2,200	2,900	33
Architecture	1,200	1,600	41
Crafts	800	900	8
Design: Product, Graphic and Fashion Design	500	1,500	238
Film, TV, Video, Radio & Photography	3,700	2,100	-42
IT, Software & Computer Services	12,800	14,100	10
Publishing	1,200	900	-28
Museums, Galleries & Libraries	1,700	1,500	-12
Music, Visual & Performing Arts	2,100	2,000	-5
DCMS Total	26,200	27,500	5
Tech Nation sectors	23,100	22,400	-3

Source: NOMIS, ONS, BOP Analysis

Figure 7. **Percentage Change in Number of Creative Industries Workers between 2010 and 2015 in GBSLEP and Comparator LEPs**

Change in Creative Industries employment 2010-15	GBSLEP (%)	Competitor LEPs (%)
Advertising & Marketing	33	44
Architecture	41	19
Crafts	8	-17
Design: Product, Graphic and Fashion Design	238	44
Film, TV, Video, Radio & Photography	-42	11
IT, Software & Computer Services	10	32
Publishing	-28	-20
Museums, Galleries & Libraries	-12	-25
Music, Visual & Performing Arts	-5	-21
DCMS total	5	13
Tech Nation sectors	-3	14

Source: ONS/NOMIS/BOP Analysis

Figure 6 shows that many of these sectors also saw employment falls over the same time period in competitor LEPs.

- Employment in Publishing; Museums, Galleries & Libraries; and Music, Visual & Performing Arts fell in both GBSLEP and comparator LEP areas.
- Crafts and Film, TV, Video, Radio & Photography pulled in different directions. Crafts employment grew in Greater Birmingham, while declining elsewhere. Film, TV, Video, Radio & Photography declined in the GBSLEP area, while growing in other LEP areas.

Overall, the rate of employment growth in the DCMS sectors in Greater Birmingham failed to keep pace with that in comparator LEP areas. This indicates that, while the Cultural and Creative Industries in Greater Birmingham are increasingly important to the wider economy of the city region and have expanded in scale and significance, it would not be unreasonable to expect these sectors to grow more quickly and match that achieved by competitor cities.

From this we conclude that **the potential of the Cultural and Creative Industries in Greater Birmingham remains unfulfilled.**

To complement our analysis of ONS data, we have also analysed LinkedIn data. We have analysed the number of creative occupations displayed on LinkedIn that are recorded as being resident in Greater Birmingham and Solihull and attached to industries recognised by LinkedIn that map on to the DCMS Cultural and Creative Industries.

As illustrated in the figure below, IT, Software & Computer Services, and Advertising & Marketing, have the most such LinkedIn profiles, underlining the importance of these sectors to Greater Birmingham.

Figure 8. **Number of Creative Occupation LinkedIn Profiles by Sector in Greater Birmingham**

IT, Software & Computer Services	10,095
Advertising and Marketing	7,452
Music, Performing & Visual arts	3,351
Publishing	2,893
Film, TV, Video, Radio & Photography	2,446
Architecture	1,764
Designer	1,546
Museums, Galleries & Libraries	452
Total	29,999

Source: LinkedIn/BOP Analysis

3.2 Creative Economy Workers

In addition to the Creative Industries, the Creative Economy also captures creative workers in the wider economy. We have analysed the creative economy in Greater Birmingham by accessing the Virtual Microdata Lab (VML) maintained by the ONS. The VML is a facility for providing secure access to sensitive detailed data, enabling more granular national statistical data to be accessed. Figure 9 (below) shows the proportion of creative workers (SOCs) within the Creative Industries (SICs) in Greater Birmingham. Therefore, for example, of the 2,900 workers in Advertising & Marketing in the GBSLEP area, 37% are employed in creative occupations – which are typically thought to be the occupations that are most innovative and generate most value.

Figure 9. **Proportion of Creative Occupations in Creative Industries**

SIC grouping:	Creative workers:
Advertising & Marketing	37%
Architecture	26%
Design and Crafts	30%
Film & TV	51%
IT	30%
Publishing and Music	17%
Museums	20%

Source: VML and BOP Consulting analysis

As Figure 10 (opposite) illustrates, a high proportion of workers in creative occupations are employed outside the Creative Industries in Greater Birmingham.

The proportion of creative workers employed outside the Creative Industries is particularly high for SOCs associated with Advertising, Architecture, Design and Crafts and Publishing/Music.⁸

Figure 10. **Proportion of Creative Occupations outside the Creative Industries**

SOC grouping:	Outside Creative Industries:
Advertising & Marketing	91%
Architecture	68%
Design and Crafts	82%
Film & TV	45%
IT	47%
Publishing and Music	76%
Museums	45%

Source: VML and BOP Consulting analysis

⁸ When undertaking analysis in the VML, we had to merge Design and Crafts and Publishing and Music to clear the disclosure thresholds enforced by the VML.

3.3 Skill levels and requirements of Creative Industries workers

Creative employers, particularly those involved in software-related activity, frequently cite unpreparedness of graduates to enter the world of work as a barrier to growth. We analysed to UK Commission for Employment and Skills (UKCES) survey data at a national and regional level to see if this could be affecting vacancy rates in creative sectors across the West Midlands. This survey was undertaken in 2015 and involved UKCES surveying 91,000 businesses across the UK on their skill requirements.

The evidence (see Figure 11, below) shows no significant difference between the preparedness for work of graduates working in the creative industries in the West Midlands and elsewhere in the UK. The table below tells us, for example, that Creative Industries businesses in the West Midlands consider 1% of graduates that they employ to be very poorly prepared for work – with creative industry businesses across the UK, similarly, considering 2% of graduates that they employ to be very poorly prepared for work.

Figure 11. Preparedness for work of University graduates in Creative Industries in West Midlands

	West Midlands	UK
Very well prepared	26%	23%
Well prepared	58%	62%
Poorly prepared	15%	13%
Very Poorly prepared	1%	2%

Source: UKCES/BOP Analysis

Figure 12 (below) shows that there are no significance differences between the West Midlands and the rest of the UK in respect of the proportionate amount of vacancies in the creative industries. 78% of such businesses in the West Midlands have no vacancies, while 76% across the UK do so. While this indicates that 22% of Creative Industries businesses have vacancies in the West Midlands, it is not possible to derive from this a total number of vacancies for the West Midlands creative industries, as the overall size of the businesses responding to the UKCES survey is not known.

However, in some cases, some quite pressing vacancy requirements appear. 7% of West Midlands Creative Industries businesses have vacancies equivalent to 10% to 19.9% of their overall staff. This indicates that around one-in-fourteen (7%) of West Midlands Creative Industries businesses are seeking to fill up to one-in-five (20%) of their job roles.

Figure 12. **Number of vacancies as a proportion of number of employees within firm in creative industries in West Midlands and UK**

Vacancies relative to total headcount of firm:	Proportion of creative businesses with this number of vacancies:	
	West Midlands	UK
0	78%	76%
>0 and <1%	0%	0%
1 - 1.99%	2%	1%
2 - 2.99%	1%	1%
3 - 3.99%	1%	1%
4 - 4.99%	2%	1%
5 - 5.99%	3%	6%
10 - 19.99%	7%	6%
20 - 29.99%	2%	3%
30 - 49.99%	1%	2%
50 - 99.99%	1%	1%
100+	0%	0%
Don't know	1%	1%

Source: UKCES/BOP Analysis

3.4 Vacancies

We have analysed LinkedIn to better understand the volume and context of current vacancies within three sectors of particular importance to Greater Birmingham: Advertising & Marketing, Design, and IT. These results are illustrated over the three figures below and we make the following observations:

- Of the three sectors analysed, the largest number of vacancies is in IT - which is consistent with this sector being the largest in Greater Birmingham, both in terms of number of workers and enterprises.
- The Design vacancies speak to the relevance of this discipline across a wide range of settings - engineering, mechanical, architectural, web.
- Marketing seems to be generating many more vacancies than Digital Marketing or Public Relations, reflecting the significant levels of employment in Advertising & Marketing outside the Creative Industries.

Figure 13. **Number of Design Vacancies by Type Advertised on LinkedIn in Birmingham**

Designer	207
Design engineer	103
Product designer	43
Design teacher	39
Design manager	29
Mechanical designer	25
Architectural design	21
Web designer	14
Graphic designer	13
Electrical design engineer	12
Total	506

Source: LinkedIn/BOP Analysis

Figure 14. **Number of IT Vacancies by Type Advertised on LinkedIn in Birmingham**

Software	991
Programmer	368
Software engineer	365
Software developer	359
Web developer	55
PHP developer	45
Frontend developer	44
Telecommunications	38
Java developer	20
IT Manager	18
IT Analyst	18
Solutions Architect	18
Lead developer	13
Android developer	12
Software specialist	10
Total	2,374

Source: LinkedIn/BOP Analysis

Figure 15. **Number of Advertising and Marketing Vacancies by Type Advertised on LinkedIn in Birmingham**

Marketing	906
Advertising	279
Content	242
Digital marketing	68
Public relations	12
Total	1,507

Source: LinkedIn/BOP Analysis

The large proportion of marketing vacancies within this correlates with our analysis of the role of Advertising & Marketing within the wider Creative Economy, which shows dispersal of 80% of those roles in companies outside the Creative Industries.

3.5 Diversity

The evidence is clear – Birmingham is the youngest city in the UK, and the commercial opportunities to exploit the diversity of its population have already been seized by other sectors, including life sciences.

But the Creative Industries have yet to fully grasp this opportunity. Although national statistics do not provide a specific breakdown of jobs by ethnicity to show the number and proportion of BAME workers in creative occupations, it is well understood that these communities are underrepresented in the Creative Economy.

Whilst this reflects a national shortcoming – as recognised in other studies, such as the British Film Institute Future Film Skills report of June

2017 - Greater Birmingham & Solihull needs to address the problem in all its dimensions: economic opportunity, representation, social cohesion and innovation.

Interventions such as RE:Present and ASTONish highlight the wider importance of engagement in culture. It isn't just about representation, important as that is: the evidence from these programmes is that the engagement in cultural activity gives participants confidence to embark on entrepreneurial activity, not limited to the creative industries.

Recent work to attract Channel 4 to the West Midlands highlights the potential for broadcasters and the media industry to exploit the diversity of Greater Birmingham and the wider West Midlands region as a 'laboratory' to address the development of new content and formats to engage not only future UK audiences, but global markets.

3.6 Future jobs growth

We would single out three areas in which trends analysis suggests there will be continuing growth in jobs:

- Data analysis and behavioural insights, already identified as a strength within the region, will become increasingly important in the context of rising adoption of artificial intelligence and machine learning
- IT, software development and computer services
- Multitalented and multiplatform content producers

These 'horizontal' capabilities will be increasingly important within existing forms of creative media and essential to the development of newer immersive forms, both for entertainment, education and other industrial applications. Demand for those skills will increase with growing understanding of, market demand for and access to 'commodity' tools to produce a range of augmented and virtual reality content and applications.

3.7 Future skill needs

The changes in production, distribution and consumer technologies that we have observed, and the increasing importance of data analysis and behavioural insights in developing new products and services, will change the nature of employment in both Creative Industries and the wider Creative Economy. This will in turn have an impact on the type of specialist skills, and combination of skills, needed in those industries.

VML data points up the dispersal of IT and Creative Services jobs across the whole economy. This, together with LinkedIn data on job vacancies, sends a strong signal about the strength of demand for skilled creative workers in Greater Birmingham.

Skills in advertising, marketing (including data analysis and digital marketing), IT and software development are transferable across the economy. The differences in their dispersal across all sectors of the economy and all parts of the LEP area reflects the difference in occupational structures between:

- Advertising & Marketing, where many small companies and (one would expect) nearly all medium and large sized companies, irrespective of industry sector, will employ ‘in house’ marketing staff as well as making use of specialist agencies. This is reflected in the fact that 80% of advertising and marketing jobs are in companies outside the creative sector.
- IT, Software & Computer Services, where the more specialist nature of the skills involved means that many more firms outside the creative and digital sectors will outsource their requirements to specialist agencies, which is reflected in the fact that four in every five IT jobs exists within the creative and digital industries.

We further analysed UKCES data to better understand the level of qualification/skill held by workers in the cultural and creative industries in the West Midlands as compared with the rest of the UK. Our findings, as illustrated in the figure below, show in 20% of such businesses in the

West Midlands, fewer than 20% of staff have Level 4 qualifications, whereas at only 14% of such businesses across the UK is this the case.

At 36 percent of cultural and creative businesses across the UK, more than 80% of staff hold Level 4 qualifications. In contrast, at only 25% of such businesses in the West Midlands is this the case.

In general, it seems, therefore, that qualification levels are lower across cultural and creative businesses in the West Midlands than across the UK. This suggests that West Midlands creative businesses may be less well equipped to deal with change and innovation than competitors in other UK cities and regions.

Of course, it is the case that many people make fantastic contributions to the creative industries without holding Level 4 qualifications, but we also take higher level qualifications as a proxy for skill and adaptability. According to this proxy, there are fewer highly skilled individuals in the jobs market in GBSLEP than in some comparator cities.

With Greater Birmingham showing lower concentrations of creative businesses than competitor LEPs – reflecting a less strongly ‘clustered’ sector, with less potential for exchange of knowledge and skills to promote innovation – this carries the risk that employers in the region will continue to struggle to identify and recruit people with relevant skills and experience for the new jobs that will emerge over the next two decades.

The final risk attached to lower levels of creative skills – particularly when combined with observed lack of scale within creative businesses in Greater Birmingham & Solihull – is an increased threat from automation of more ‘process driven’ jobs within the sector.

Figure 16. **Number of staff qualified to Level 4 or above as a**

proportion of number of employees

	West Midlands	UK
Fewer than 20%	20%	14%
20% to 49%	20%	16%
50% to 80%	28%	27%
More than 80%	25%	36%
Don't know	7%	7%

Source: UKCES/BOP Analysis

4. Value and Significance of Creative Economy

This chapter sets out of assessment of the contribution to Gross Value Added (GVA) of the creative and cultural industries to GBSLEP. We also assess value in terms of cluster strength, through both Location Quotient (LQ) analysis and analysis of alternative data sources, including LinkedIn and Meetup.

4.1 GVA of Creative Industries in Greater Birmingham

Having reported upon the levels of employment in Greater Birmingham and Solihull in our previous chapter, we calculate the Gross Value Added (GVA) associated with each of these sectors by multiplying the number of workers by the GVA-per-worker for each of these sectors as reported in the most recent version of the Annual Business Survey (ABS) published by the Office of National Statistics (ONS).

The results of this analysis are illustrated in the figure below and show the relative economic importance of each of the creative sectors.

Figure 17. **GVA of Creative Industries in GBSLEP in 2015 (£m)**

Creative industries grouping	GVA (£m)	% of total
Advertising and Marketing	327	15%
Architecture	101	5%
Crafts	42	2%
Design: Product, Graphic and Fashion Design	106	5%
Film, TV, Video, Radio and Photography	191	9%
IT, Software, and Computer Services	1,184	56%
Publishing	59	3%
Music, Visual, Performing Arts	106	5%
DCMS Total	2,116	100%
Tech Nation sectors	2,337	n/a

Source: NOMIS and Annual Business Survey (ABS) with BOP Consulting analysis

4.2 Creative employment in Greater Birmingham & Solihull

Analysis in the VML allowed us to estimate the number of creative workers inside and outside the creative industries. We combined these proportions with the employment numbers contained in NOMIS to estimate the employment sustained by the creative industries and creative economy in Greater Birmingham and Solihull. By assuming that the GVA-per-worker generated by creative workers outside the creative industries is equal to the GVA-per-worker generated per worker in the Greater Birmingham and Solihull creative industries, we have estimated the GVA of the Greater Birmingham and Solihull creative economy. We present these results on the employment and GVA of the creative industries and creative economy of Greater Birmingham and Solihull in the figures below.

Figure 18. **Employment in Creative Industries and Creative Economy in Greater Birmingham and Solihull (2015)**

	Creative Industries Employment	Creative Economy Employment
Advertising and Marketing	2,900	14,350
Architecture	1,600	2,500
Crafts	900	2,050
Design	1,500	3,500
Film and TV	2,100	3,000
IT	14,100	17,900
Museums	1,500	1,700
Music	2,000	3,100
Publishing	900	1,350
Total	27,550	49,600

Source: VML, NOMIS and BOP Consulting

Figure 24. **GVA of Creative Industries and Creative Economy in Greater Birmingham and Solihull (2015)**

Industry grouping	Creative Industry GVA (£m)	Creative Economy GVA
Advertising and Marketing	327	1,617
Architecture	101	156
Crafts	42	98
Design	106	244
Film, TV, Video, Radio and Photography	191	271
IT	1,184	1,503
Music	106	163
Publishing	59	91
Total	2,116	4,144

Source: VML, NOMIS and BOP Consulting

4.3 Significance of Creative Economy to GBSLEP Economy

The table below shows the scale of the Creative Economy compared to key employment sectors in Greater Birmingham and Solihull. The employment footprint of the Creative Economy is very nearly equivalent to that of construction – with the Creative Economy being a more important source of employment than other key sectors including: AME; Transport & Logistics; ICT/Digital & Tech; and Mid-Tech Manufacture.

Figure 25. **Employment in Creative Economy in Greater Birmingham and Solihull compared to key Employment Sectors**

	Number of jobs	% of GBSLEP total
BPFS	193,500	21.9%
Food & drink	69,400	7.8%
Tourism/Visitor Economy	61,700	7.0%
Construction	50,300	5.7%
Creative Economy	49,600	5.6%
AME	36,500	4.1%
Transport & Logistics	29,500	3.3%
ICT/Digital & Tech	21,800	2.5%
Med tech manufacture	2,100	0.2%

Source: BRES, VML, NOMIS and BOP Consulting

4.4 Creative originals, content, services, experiences

We have looked at our data results in another way to bring new insight to the strengths of the Creative Economy and Creative Industries in Greater Birmingham. This was done by subdividing the Creative Industries into groups reflecting their outputs and value chains:

- Creative Content – the audiovisual and publishing sectors, production and distribution of entertainment content in either physical, broadcast or electronic form
- Creative Experiences – live events, education, museums, galleries and libraries
- Creative Originals – design and production of artefacts, including jewellery and fashion
- Creative Services – advertising, marketing, architecture, graphic design, translation services and other publishing

This subdivision maps on to Creative Industries SIC codes as follows:

Figure 26. **Allocation of Creative SICs to Creative Groupings**

Creative SICs:	Proportion of SICs allocated to groupings:			
	Content	Experience	Original	Service
Public relations and communication activities				100
Advertising agencies				100
Media representation				100
Architectural activities				100
Manufacture of jewellery and related articles			100	
Specialised design activities			67	33
Motion picture, video and television programme production activities	100			

Motion picture, video and television programme post-production	100			
Motion picture, video and television programme distribution	100			
Motion picture projection activities	100			
Radio broadcasting	100			
Television programming and broadcasting activities	100			
Photographic activities			100	
Publishing of computer games	100			
Other software publishing	100			
Computer programming activities	100			
Computer consultancy activities	100			
Book publishing	100			
Publishing of directories and mailing lists				100
Publishing of newspapers	100			
Publishing of journals and periodicals	100			
Other publishing activities	50			50
Translation and interpretation activities				100
Library and archive activities		100		
Museum activities		100		
Sound recording and music publishing activities	100			
Cultural education		100		
Performing arts		100		
Support activities to performing arts		100		
Artistic creation			100	
Operation of arts facilities		100		

Source: BOP Consulting Analysis

When re-grouped as above, our results on employment and GVA within the creative industries and creative economy of Greater Birmingham and Solihull are as per the figures below.

Figure 27. **Employment by Creative Content/ Experience/ Original/ Service Grouping (2015)**

	Content	Experience	Original	Service
Creative Industries	16,850	2,900	2,500	5,300
Creative Economy	22,000	4,000	5,300	18,400

Source: VML/NOMIS/BOP Consulting analysis

Figure 28. **GVA by Content/ Experience/ Original/ Service Grouping (£m, 2015)**

	Content	Experience	Original	Service
Creative Industries	1,439	48	153	476
Creative Economy	1,859	89	321	1,875

Source: VML/NOMIS/BOP Consulting analysis

This analysis emphasises the scale, significance and value of the Creative Content production sector to the Creative Industries in Greater Birmingham: despite recent falls in employment, it still accounts for more jobs (16,850) and output (£1.4 billion) than any other of the creative groupings analysed in this way.

It also emphasises the significance of Creative Services in the wider economy, with an output of £1.9 billion GVA per annum.

4.5 Cluster analysis of Creative Industries in Greater Birmingham and Solihull

This section analyses cluster strength in Greater Birmingham and Solihull through analysis of both NOMIS and Meetup data.

4.5.1 Location Quotient (LQ) Analysis

Location quotient (LQ) is a way of quantifying how concentrated a particular industry, occupation, or demographic group is in a region as compared to the nation. It is the standard technique for measuring the strength of a particular economic cluster. In this case, we have assessed employment by creative sectors in Greater Birmingham and Solihull next to the UK as a whole. LQ scores of above 1 are indicative of clustering.

Analysis of the strength of clusters by LQ gives an indication of the relative strength of those industries in Greater Birmingham & Solihull against the national average, and highlights strengths and weaknesses. This may help focus future business support and investment.

- Figure 31 (below) indicates that there are clusters in Greater Birmingham and Solihull in Crafts; Design: Product, Graphic and Fashion Design; and IT, Software and Computer Services.
- While IT, Software and Computer Services only just clears the threshold of 1.0 associated with clustering, the case that there is a digital cluster locally is strengthened by repeating the analysis on the Tech Nation sectors and deriving a score of 1.5.

We repeat the LQ analysis at the level of the local authorities that compose GBSLEP and present our results below. For the most part, we find the same sectors that appear at the city region level (Crafts; Design: Product, Graphic and Fashion Design; IT, Software and Computer Services) associated with particular local authorities but we also note that:

- Clustering is reasonably well dispersed across Greater Birmingham and Solihull.

- Only Tamworth, Redditch and Wyre Forest lack creative clusters.
- The Design and IT clusters are more widely geographically dispersed than the Crafts cluster, which is very concentrated upon Birmingham itself - and most likely, attributable to the Jewellery Quarter.
- We do see some different kinds of clusters appearing at this more local level, i.e. Architecture in Lichfield. Even where there is a comparatively small concentration of businesses in a local area, an LQ equal to or greater than 1.0 is a useful indicator of a strength that could potentially be built upon.

Figure 29. **Creative Industries LQs in local authorities of GBSLEP (2015)**

Advertising & Marketing	0.7
Architecture	0.7
Crafts	6.2
Design: Product, Graphic and Fashion Design	1.1
Film, TV, Video, Radio & Photography	0.6
IT, Software & Computer Services	1.0
Publishing	0.2
Museums, Galleries & Libraries	0.8
Music, Visual & Performing Arts	0.6
All GBSLEP Creative Industries sectors	0.8
Tech Nation Digital Industries sectors	1.5

Source: NOMIS with BOP Consulting analysis

Figure 30. **Location Quotients indicating clusters of Creative Industries businesses in local authorities of GBSLEP (2015)**

Birmingham	
Crafts	11.8
Design: Product, Graphic and Fashion Design	1.1
Bromsgrove	
Design: Product, Graphic and Fashion Design	1.9
IT, Software, and Computer Services	1.1
Cannock Chase	
Design: Product, Graphic and Fashion Design	1.2
East Staffordshire	
Design: Product, Graphic and Fashion Design	2.2
IT, Software, and Computer Services	1.0
Lichfield	
Architecture	1.1
Crafts	1.8
Solihull	
Advertising and Marketing	1.1
IT, Software, and Computer Services	2.5
Museums, Galleries and Libraries	1.4

Source: NOMIS with BOP Consulting analysis

4.5.2 Meetup Analysis

Meetup (www.meetup.com) is a social networking site, that facilitates offline meetings in many localities across the world. Users can join groups based on various interests. The site charges organisers a fee which is based on group size and the number of organisers.

- We analysed a 25-mile radius of Birmingham but excluded those meet ups which are obviously outside of the GBSLEP area (e.g. Leamington, Warwickshire).
- We have analysed meet ups tagged as 'Tech' and 'Creative'. We browsed all relevant meetups to identify any other relevant groups which do not include these terms specifically (e.g. particular programming languages).
- We developed a definition of active groups. Active is defined as having had a meetup in the last 2 months and/or at least one meeting planned with date set.

Findings

There are 24 active technology and creative meetup groups in the GBSLEP area. 21 of these are in the Birmingham area. 22 are technology focused and two are classified as 'creative'. A further breakdown of the type of groups is displayed in the chart below.

The largest group is 'Badego: Creative Meetup Birmingham' which has 2,437⁹ members, followed by 'Brum Tech Drinks' with 2095 members. 'Tech Wednesday' (1,834 members) and 'Hacks and Hackers Birmingham' (1,060 members) are third and fourth largest respectively.

14 of the groups meet at least monthly. There are ten long-standing groups which have held over 40 meetings each. The most established are:

- **Birmingham Code**, which meets weekly and has met 150 times since it was established on Meetup.com in March 2014.
- **fizzPOP** (Birmingham's Makerspace) has held 104 weekly events.
- **West Midlands Ruby Group** has met 87 times since its inception in January 2010
- **Tech Wednesday** has held 82 meeting since February 2010.

There is one group, the Birmingham User Experience Group, which meets in Solihull. This group is not included in the map at Figure 24 (below).

Badego meets regularly in two different venues within the centre of Birmingham.

⁹ All numbers are accurate to 28 June 2017

Figure 31. **Classification of tech and creative meetups in Greater Birmingham and Solihull**

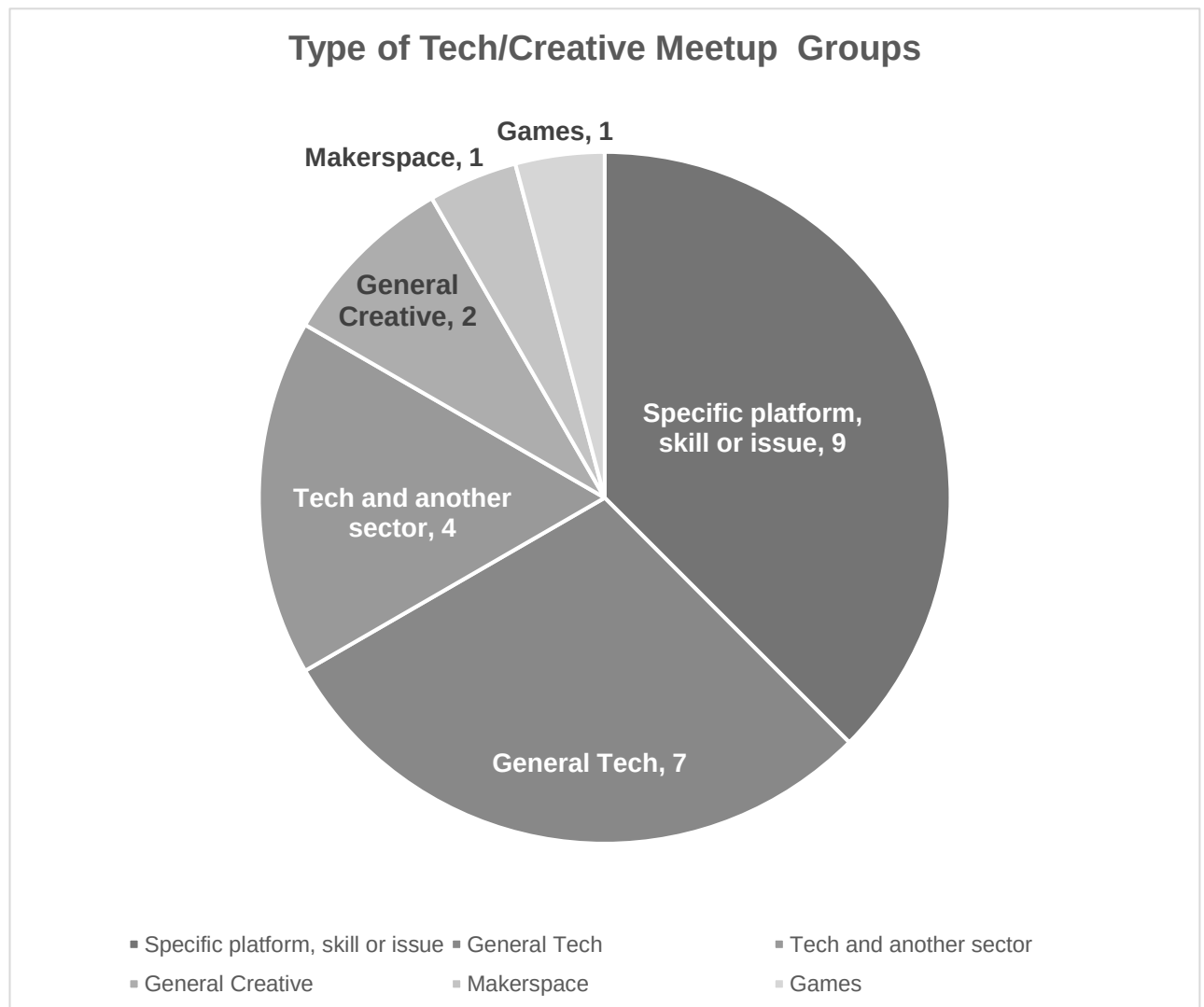
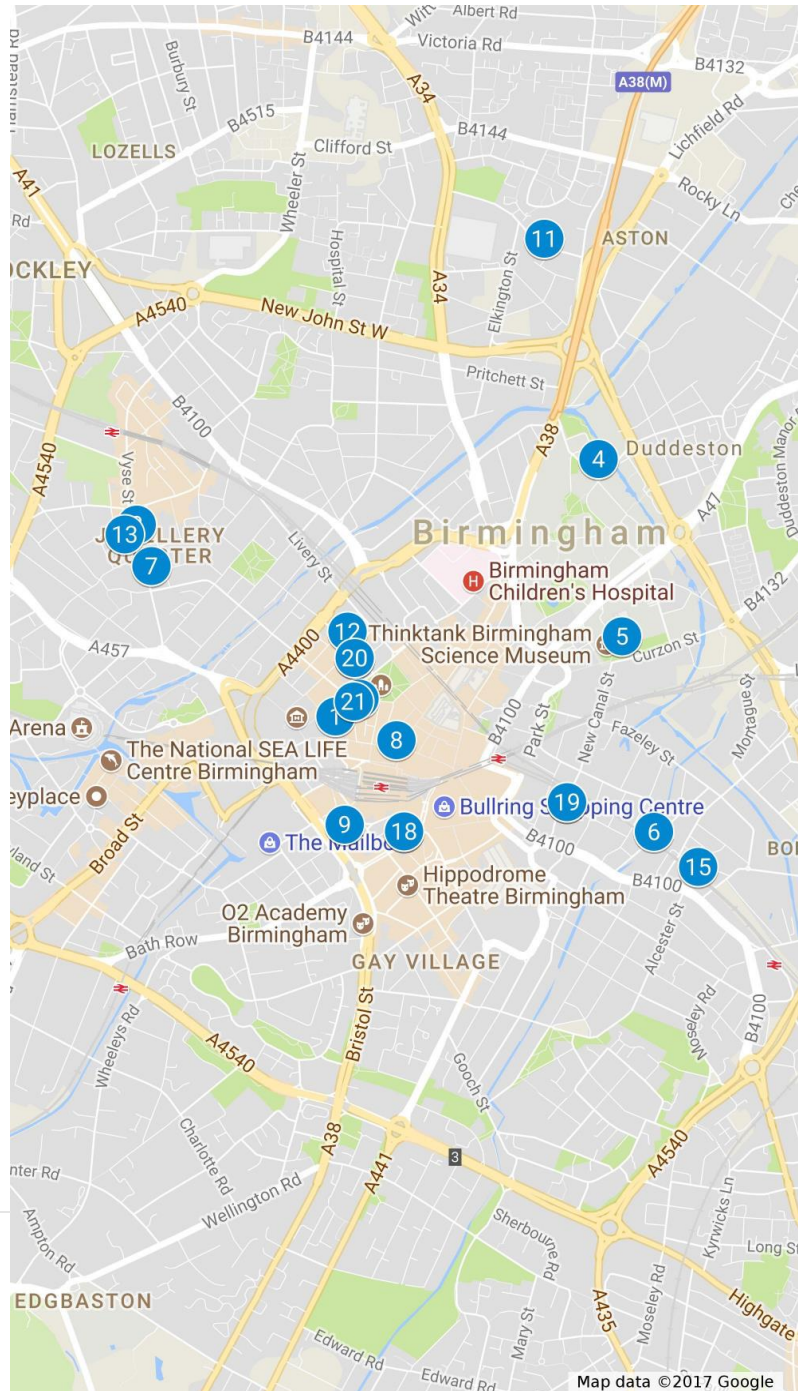


Figure 32. Location of tech and creative meetups in Greater Birmingham and Solihull

Birmingham Creative/ Digital Meetup groups

Meetup Data

- 1 Badego - Creative Meetup
- 2 Badego - Creative Meetup
- 3 Brum tech drinks
- 4 Tech Wednesday
- 5 Hacks and Hackers
- 6 fizzPOP - Makerspace
- 7 HydraHack
- 8 Fusion Meetup
- 9 Wordpress Meet up
- 10 Birmingham Open Code
- 11 Microsoft Data Platform
- 12 Lean Agile Birmingham
- 13 West Midlands Ruby Group
- 14 Birmingham Xamarin Mobile Cross Platform User Group
- 15 Glug Birmingham
- 16 Birmingham Digital and DevOps Meetup
- 17 Birmingham User Experience Group
- 18 Open Rights Group Birmingham
- 19 NetSquared Midlands
- 20 Hackers & Founders
- 21 Birmingham Videogame Indie Developers



5. The competitive position of GBSLEP

This chapter reviews the performance of the creative industries in Greater Birmingham and Solihull against comparator LEPs by analysing and comparing:

- The number of creative enterprises: both in absolute terms and as a proportion of local enterprise bases.
- The number of creative industry workers: both in absolute terms and as a proportion of local workforces.
- Location Quotient (LQ) scores for the creative industries within these LEPs.

We have then undertaken more granular LQ analysis at the level of each creative industries sector to identify the creative clusters that exist, at least in LQ terms, within these areas. We see that clustering seems to correlate with higher numbers of creative workers and enterprises.

We find that while Greater Birmingham and Solihull outperforms a number of other geographies on a range of indicators, performance elsewhere exceeds that of Greater Birmingham and Solihull, suggesting some degree of unfulfilled potential in Greater Birmingham and Solihull.

In the figure below, we rank LEPs according to the proportion of local enterprises that operate within the creative industries, as defined by the Department of Digital, Culture, Media and Sport (DCMS). We see that Greater Birmingham and Solihull outperforms a number of LEPs on this indicator but lags beyond others, in particular Oxfordshire, West of England (including Bristol), and Greater Cambridge and Greater Peterborough.

Figure 33. **Creative Enterprises in Relevant LEPs by Number and as Proportion of Total Enterprises (2015)**

LEP areas	DCMS Enterprises:	As % of total enterprises:
Oxfordshire	3,810	12%
West of England	5,400	12%
Greater Cambridge and Greater Peterborough	5,875	10%
Coventry and Warwickshire	3,220	9%
Leeds City Region	7,845	7%
Greater Manchester	7,355	7%
Greater Birmingham and Solihull	5,135	7%
Derby, Derbyshire, Nottingham and Nottinghamshire	4,730	7%
Liverpool City Region	2,675	6%
North East	3,145	6%
Sheffield City Region	3,350	6%
Black Country	1,620	5%

Source: NOMIS and BOP Analysis

If Greater Birmingham and Solihull were to match the performance of Oxfordshire on number of enterprises, it would add 3,965 enterprises to the business base of Greater Birmingham and Solihull.¹⁰

Analysis of creative workers presents a similar picture. Again, we find that Greater Birmingham and Solihull outperforms a number of LEPs in terms of the number of creative industry workers and the proportion of the local workforce that they constitute. Equally, however, Oxfordshire, West of England, and Greater Cambridge and Greater Peterborough are the three strongest performers.

If Greater Birmingham's Creative Industries were to match the performance of those in the strongest LEP (Oxfordshire), they would add just over 30,000 jobs to the GBSLEP area.¹¹

Figure 34. **Creative Workers in Relevant LEPs by Number and as Proportion of Total Employment (2015)**

	DCMS Employment:	As % of total workforce:
Oxfordshire	22,977	6%
West of England	23,088	4%
Greater Cambridge and Greater Peterborough	28,119	4%
Greater Manchester	44,570	4%
Coventry and Warwickshire	14,430	3%
Greater Birmingham and Solihull	27,547	3%
Leeds City Region	44,543	3%
North East	23,889	3%
Liverpool City Region	14,571	2%
Derby, Derbyshire, Nottingham and Nottinghamshire	21,817	2%
Sheffield City Region	16,241	2%
Black Country	6,823	1%

Source: NOMIS and BOP Analysis

¹⁰ This is the total number of enterprises in Greater Birmingham and Solihull multiplied by the proportion of creative businesses in Oxfordshire minus the number of such businesses already in Greater Birmingham and Solihull.

¹¹ This is the total number of workers in Greater Birmingham and Solihull multiplied by the proportion of creative industry workers in Oxfordshire minus the number of such workers already in Greater Birmingham and Solihull.

Below we see the strong performance of Oxfordshire, West of England, and Greater Cambridge and Greater Peterborough carried into their Location Quotient performance for their creative industries as a whole. On this measure, they are the only geographies with LQs above 1.0 – indicating that they can be considered to be sustaining creative clusters across the creative industries as a whole.

Figure 35. **Location Quotients for Creative Industries by LEP (2015)**

Oxfordshire	1.6
West of England	1.0
Greater Cambridge and Greater Peterborough	1.0
Greater Manchester	0.9
Coventry and Warwickshire	0.8
Greater Birmingham and Solihull	0.8
Leeds City Region	0.8
North East	0.7
Liverpool City Region	0.5
Derby, Derbyshire, Nottingham and Nottinghamshire	0.5
Sheffield City Region	0.5
Black Country	0.4

Source: NOMIS and BOP Analysis

Rather than having clusters across the creative industries as a whole, it is more usual to think about clusters within the particular creative sectors. This is because shared specialisms, expertise and agglomeration potential are likely to be stronger within, rather than across, creative sectors. Figure 33 (below) shows that creative clusters are quite widely dispersed across the LEPs that we have analysed.

These clusters demonstrate, and are sustained by, local strength and specialism. Building upon their particular strengths is likely to be important to the future growth of the creative industries within these geographies.

We note that the crafts cluster in Greater Birmingham and Solihull – with an LQ of 6.2 – is the strongest of any of the clusters identified in our analysis.

Figure 36. **Clusters by LEP as evidenced by Location Quotients (2015)**

Advertising & Marketing	
Oxfordshire	2.4
Leeds City Region	1.5
Architecture	
West of England	2.0
Crafts	
Greater Birmingham and Solihull	6.2
Greater Cambridge and Greater Peterborough	2.7
Sheffield City Region	2.3
Design: Product, Graphic and Fashion Design	
Coventry and Warwickshire	1.4
Oxfordshire	1.3
Derby, Derbyshire, Nottingham and Nottinghamshire	1.2
Greater Birmingham and Solihull	1.1
Leeds City Region	1.0

Film, TV, Video, Radio & Photography	
Greater Manchester	1.2
IT, Software & Computer Services	
Oxfordshire	1.3
Coventry and Warwickshire	1.2
Greater Cambridge and Greater Peterborough	1.2
Museums, Galleries & Libraries	
North East	1.3
Leeds City Region	1.2
Oxfordshire	1.0
Liverpool City Region	1.0
Greater Cambridge and Greater Peterborough	1.0
Publishing	
Oxfordshire	4.7
Greater Cambridge and Greater Peterborough	1.7
West of England	1.1

6. Findings from Consultation

Three consultation workshops, held over a period of a week in July 2017, were designed to allow different parts of the cultural and commercial creative sector, and employers and educators within the wider Creative Economy, reflect on the evidence gathered in the initial stages of research. They provided a forum for individual artists, arts organisations both large and small, commercial creative businesses and funders to express their view of the cultural and economic ecology or of the Creative Industries sector in Greater Birmingham, highlight structural barriers to growth and suggest ways in which the public sector could intervene more effectively to catalyse and underpin growth of the Creative Industries.

BOP presented a number of potential scenarios, tailored toward the audience for each of the workshops, that drew out key findings of initial research.

Key findings from consultation were:

- Evidence of spillovers between cultural organisations and the commercial creative sector – including innovative and international business models of arts organisations such as Ikon Gallery, with its large international specialist publishing business, and Stan’s Café, whose innovative and scalable theatre show, “Of All The People In All The World” has been licensed as a format to other producers in New York, Melbourne, Buenos Aires and across Europe since its original production in 2003.
- Leadership is not confined to large cultural organisations, nor to publicly funded ones. Birmingham Hippodrome is a commercial operation that plays a major role in offering greater access to its building and promoting diversity in cultural leadership and entrepreneurship amongst its neighbouring communities. Smaller festivals such as Flatpack and independent organisations such

as Ikon and Stan’s Café act as intermediaries and are diversifying their business model to reduce dependency on public subsidy, using a mix of subsidised and commercial activity to identify and develop new talent and help identify funding to secure and exploit IP created in the process. This expertise and knowledge is being transferred directly and indirectly to other creative practitioners and firms, through festivals, training programmes and workshops.

- Not all structural barriers to growth were located within the Creative Industries; several participants commented that inadequate transport infrastructure acted as a disincentive to audiences, who may be unable to travel home by public transport after shows; and how the road system served more to divide creative communities than to act as a means of communication between them. When applied to analysis of the dispersal of Creative Industries sectors and employment across GBSLEP geographies, it is possible to see some correlation between the reported transport barriers and the lower levels of clustering in towns and local authorities furthest from the city core.
- There was a shared view that creative organisations needed to take the initiative in their relationship with public bodies, particularly in asserting their views on how changes in the public realm could best be turned to the advantage of the creative sector before, during and after development. Suggestions included taking over hoardings for art and promotional activity to raise awareness of cultural organisations in the vicinity during redevelopment work; and ensuring effective wayfaring and signage of venues.

The table below highlights some of the most pertinent comments from the workshop sessions.

Figure 37. **Highlighted comments from consultation workshops**

Consultation topic	Workshop 1 – arts and cultural organisations	Workshop 2 – Digital & Creative Industries	Workshop 3 – Innovation system partners
Space and infrastructure	Lack of affordable workspace. Include cultural provision in plans for new districts. Geography and public transport present barriers; huge transport blockages, North, South, East, West.	Lack of Internet Exchange. Need a 5G testbed and integrated fixed line/5G plans for new developments. Collaboration space for bigger companies and SMEs.	Hard for artists and not-for-profit businesses to find suitable spaces – City Council should sell freehold to unused properties for £1. Make space for art on public billboards and hoardings around new developments.
Markets	Develop ‘anchor’ shows and repeatable business (eg for international touring).	Corporate ‘labs’ needed to help large and small companies explore new markets.	Greater response from intermediaries like Creative England to help brokerage.
Supportive environment	Businesses need to take responsibility for their own development.	Put culture and community first in new developments to trigger social impacts.	Support for ‘LARC’-type model for joint action between arts orgs.
Access to finance	National funding schemes are not delivering for Birmingham.	Large number of start-ups, but low number of scale-ups.	Help SMEs learn the language of financing applications.
Human capital and talent	Change make-up of boards of cultural organisations to trigger diversity.	Disposable income per capita in London is £66pcm; in Birmingham, £1,000 pcm.	Greater Birmingham offers a perfect platform for inclusive growth.
Culture of the cluster	Create a shared open data resource – ‘cultural dashboard’.	Large businesses – inward investors especially – are not engaging with the sector.	Creative Industries sectors are ‘siloed’ – there is no overarching community.

7. SWOT of GBSLEP Creative Economy

This chapter draws upon insights from our data analysis, interviews and workshops, and our literature review to present this analysis of the Strengths, Weaknesses, Opportunities and Threats (SWOT) of the GBSLEP creative economy.

SWOT analysis is undertaken by organisations to identify their internal strengths and weaknesses, as well as their external opportunities and threats. In the case of Greater Birmingham, success can be conceived in a variety of ways, such as:

- The scale of the **Creative Industries**– its GVA and FTE employment contributions.
- The significance of these contributions within the overall GVA and FTE employment contributions of Greater Birmingham.
- The importance of these contributions to the achievement of wider economic goals set by relevant strategic economic plans, i.e. those developed by GBSLEP and the WMCA.

At the level of the wider economy, this relates to economic sectors beyond the Creative Industries. How might, for example, better ways of working between the Creative Industries and other sectors grow the output of the Creative Economy?

- The scale and dispersal of workers in creative occupations across the economy as a whole – its GVA and FTE employment contributions.
- How that measure of a wider **Creative Economy** - both the creative industries and creative employment in the wider economy - helps us understand where creativity is core to the competitiveness and growth of other industry sectors

Within Greater Birmingham and across the West Midlands, opportunities exist within Business, Financial and Professional Services and fusions of creative disciplines and traditional manufacturing strength to create new forms of digital manufacturing.

At the geographic level, we are concerned with how the benefits of growth in the Creative Industries and the Creative Economy are dispersed across Greater Birmingham. Where and in what sectors does it demonstrate nationally significant concentrations of businesses and skills? Is there evidence of clustering across local authority boundaries? What message does the aggregation of the creative talent pool at a LEP and city regional level send to potential investors?

In turn, external opportunities and threats can be conceived of at sectoral and geographic levels. How can events beyond Greater Birmingham create opportunities and threats for the Creative Economy within the LEP area? How can, for example, policy in other city regions be thought likely to heighten competitive pressure on the Creative Economy within Greater Birmingham? Many city regions are becoming more ambitious for their creative economies, so Greater Birmingham must strive to be as competitive as possible. How significant is the Creative Economy within the West Midlands Combined Authority?

Figure 38. **SWOT analysis of creative economy in Greater Birmingham & Solihull**

Strengths	Weaknesses
• ICT, Software & Computer Services sector is nationally competitive – particularly when looking at the wider ‘Tech Nation’ definition of digital industries • Unheralded strengths in advertising and marketing, built on excellence in data analysis and behavioural insight (accentuated at the Combined Authority level) at universities across the West Midlands • Craft sector is intensively clustered and growing across the city region • Design growing at a phenomenal rate and well dispersed • Large hinterland presents great opportunity to advertising and marketing businesses, reflected in the scale of their contribution • Some well-coordinated and creatively driven propositions around skills, both formal and informal: Creative Alliance, BOA, BMet Digital & Creative Career College, Quickcode Labs, • Investment in innovation by Universities – e.g., STEAMhouse • Great transport connectivity – at the centre of national road and rail networks • Much lower accommodation costs than other competitor creative cities (eg London, Bristol, Oxford, Cambridge) – cheaper to set up and run a business, and sufficient supply of different types of housing at affordable prices to make it possible for young people to purchase a home near to their place of work.	• Uneven growth between sectors and across geographies • Lack of large businesses, and barriers to ‘scale up’ • The city region’s self-image is defined by a lack of self-confidence amongst creative businesses and cultural organisations • Supply of skills at Level 4 and above well below national average • The diversity of the city region’s people is not reflected by or taken advantage of in its Cultural and Creative Industries • Transport and communications within and across the LEP area are less coordinated than the national and trunk routes in and out

Opportunities	Threats
• Dispersal of advertising and marketing roles geographically and across all sectors of the GBSLEP economy presents opportunities for inward investors, professionals looking to relocate and new entrants – a pool of talent for employers and a range of roles and career options for marketing and communications professionals • The statistical record stops at the point where anecdotal evidence shows a revival in high end content production, centred on the city of Birmingham but dispersed across Solihull and the city region, with potential for rebirth of the commercial sector, boosted by release of £2.1m in West Midlands Production Finance • Looking at Digital & Creative industries on a WMCA basis can help turn some 'good' investment propositions into 'great' examples of smart specialisation and internationally competitive clusters • Raising awareness of great original IP and business models from arts organisations to increase confidence amongst GBSLEP creatives = reducing dependency on public funding • Untapped growth potential in 'outer' local authorities, where Creative Industries clusters are beginning to establish themselves • Diversity – much talked about, but needs specific policy and better targeted interventions to harness benefits of Creative Industries growth for the whole population of the city region • Integrated strategy for transport/housing/culture/liveability could point up the 'quality of life' benefits for both existing residents and incomers.	• In an increasingly straitened investment climate, there is a risk that the city region's lack of image as a creative hub and scarcity of examples of businesses which have successfully 'scaled' will hinder attempts to attract new businesses • No strategy or policy in place to deal with disruption and displacement brought about by large scale plans for new infrastructure and commercial pressure on affordable workspace in existing clusters • HS2 may accelerate 'brain drain' to London, accentuate shortages of key skills and inflate salaries • Restrictions on freedom of movement as a result of Brexit may further reduce access to talent, accentuate skills gaps and shortages, and inflate salaries as Digital & Creative industries employers in Greater Birmingham and across the UK compete for a narrower pool of skilled employees • Post-Brexit tariff and non-tariff barriers may limit growth of exports of to Europe, and make overseas touring more expensive and difficult. • Compliance with multiple regulatory regimes post-Brexit (eg, GDPR, and conflicting requirements for managing customers' data in a cross-border environment, particularly with US platforms) could increase costs and risks for IT, advertising and marketing businesses.

7.1 Strengths of GBSLEP Creative Economy

The Creative Industries are increasing in economic importance to Greater Birmingham and Solihull:

- 26% increase (to over 5,100) in the number of Creative Industries enterprises within the LEP between 2010 and 2015.
- Number of workers in Greater Birmingham's Creative Industries increased by 5%, to over 27,500, over same time period.
- Creative Industries enterprises generated output of £2.1 billion of GVA in 2015.
- Adding creative employment outside the Creative Industries gives a combined creative economy of 49,900 employees – some 5.6% of the total workforce in GBS LEP – generating an output of £4.1 billion GVA per annum.

Some sectors are particularly significant in scale and concentration:

7.1.1 Advertising & Marketing

Many small companies and nearly all medium and large sized companies, irrespective of sector, have their own marketing function. It should not be surprising, then, to find that Advertising & Marketing is the creative occupation most strongly represented outside the Creative Industries. Of over 14,000 Advertising, Marketing and PR professionals in the GBSLEP area, 11,500 – 80% of the total - work in the wider, 'non-creative' economy. High levels of individual output – average GVA of £112,600 per worker – meant that this was also the most productive sector of the Creative Economy within the GBSLEP area, generating £1.6 billion in GVA in 2015.

While Advertising & Marketing agencies are not as concentrated a cluster within Greater Birmingham as Crafts, Design and IT, they employed 2,900 staff within 485 agencies and generated £327m of GVA

in 2015 – more than any sector apart from IT, Software & Computer Services.

The sense that Creative Services form a cluster of regional significance is reinforced by analysis of neighbouring LEP areas. The combined number of Advertising & Marketing employees in the Black Country and Coventry & Warwickshire rose from 500 to over 1,000 over the same period.

The extent of clustering in Greater Birmingham and Solihull compares favourably to other core cities

7.1.2 IT, Software & Computer Services

There were 14,000 workers in this category in Greater Birmingham and Solihull in 2015, representing 51% of all jobs within the Creative Industries.

The IT sector was the source of 56% of Creative Industries GVA (£1.2 billion) in Greater Birmingham in 2015, with a further 3,800 IT employees in non-creative businesses generating an additional £320m in GVA.

Boosted by the presence of specialist activity such as the games cluster focused on Leamington Spa, IT and software development is a 25,000-strong cluster across the West Midlands, with a further 8,400 employees in the sector in Coventry & Warwickshire and 2,600 in the Black Country.

7.1.3 Design: Product, Graphic and Fashion Design

Employment in this sector has increased by 238% - to 1,500 jobs - between 2010 and 2015 in Greater Birmingham and Solihull.

The design cluster is dispersed across Greater Birmingham and Solihull, with concentrations of design businesses above the national average in Birmingham, Bromsgrove, Cannock Chase and East Staffordshire. Another sector with emerging regional strengths, the number of designers has also increased in the Black Country LEP (to over 300 employees, a 245% increase) and Coventry and Warwickshire LEP (1,000 jobs, an increase of 239%) between 2010 and 2015,

As might be expected in a region with such strengths in manufacturing and product development, the majority of Design employment in Greater Birmingham still occurs outside the Creative Industries, with 57% of design employment and GVA (2,000 jobs and £139m per annum) coming from other sectors.

7.1.4 Crafts

As the heritage of the Jewellery Quarter might lead us to expect, there is very strong statistical evidence of a cluster in Birmingham, as measured by the location quotient (LQ) of the Crafts sector of over 11; in other words, jewellery designers and makers are eleven times more concentrated within the city than the national average. It may be more surprising to learn that this cluster is also dispersed across the wider Greater Birmingham and Solihull area, with evidence of clustering in Lichfield. Taking into account the wider range of craft activity across the Creative Economy, there were over 2,000 craft workers in the GBSLEP area in 2015, generating output of £97m GVA.

7.1.5 Clusters

Clusters are geographic concentrations of workers and firms, achieving higher productivity through proximity and international renown. There is evidence of clustering across Greater Birmingham & Solihull in:

- **Crafts** – particularly the design and manufacture of jewellery – represents a very strongly cluster in Birmingham, with some statistical evidence of a cluster in Lichfield.
- **Design** (product design, fashion design and graphic design) is clustered in Birmingham, Bromsgrove, Cannock Chase, East Staffordshire and Redditch – no other cluster is so widely dispersed across GBSLEP geography (and also features in Coventry & Warwickshire).
- Taking into account the nationally and internationally significant games cluster in and around Leamington Spa, Greater

Birmingham forms part of a city regional cluster of **IT, software and computer services** businesses and digital talent, taking in Birmingham, Bromsgrove, East Staffordshire and Solihull – as well as Coventry & Warwickshire.

- In addition, Lichfield evidences a small cluster in **Architecture**.

In analysing the comparative performance of the Creative Industries in Greater Birmingham against comparator LEP areas, we see:

- Creative Industries firms represent a lower percentage of the business base and are less strongly concentrated within GBS LEP than in other knowledge-intensive clusters around Oxford, Cambridge and Bristol
- Greater Birmingham has lower levels of creative employment than other core cities such as Manchester and Leeds
- Other than Crafts and Design, no other Creative Industries sector is concentrated above the national average in Greater Birmingham. This is significant because sectors that demonstrate stronger clustering benefit from greater knowledge sharing between businesses and support a stronger local jobs market
- The opportunity for further, inclusive growth through Creative Industries has not been realised in towns and local authorities such as Tamworth and Wyre Forest.

Recent discussions around the potential relocation of Channel 4 highlight that the **high-end production sector** is at an inflection point. Employment in some core content creation sectors has contracted over the period 2010-2015, although it still supports 22,000 jobs across the creative economy of Greater Birmingham. There is evidence of new investment, including that by BBC, but television, film, games and digital content production all require sustained focus and continuing investment to ensure their continuing competitiveness

7.1.6 Strength of the region's arts and cultural organisations

A strong city-wide cultural offer

Birmingham has a city centre cultural offer that is among the strongest in English cities outside London. This includes the City of Birmingham Symphony Orchestra, Birmingham Repertory Theatre, Birmingham Royal Ballet and the Ikon Gallery.

Birmingham's cultural provision is shaped by significant, world-leading institutions. Of the 38 National Portfolio and Major Partner Museum organisations across the West Midlands Combined Authority area, 24 are in Birmingham.

The institutions are popular. The Birmingham Royal Ballet annually attracts audiences in excess of 150,000 every year. The Birmingham Symphony Hall and the Birmingham Town Hall annually attract over 500,000 visitors, with Birmingham Museum & Art Gallery attracting close to 1 million in 2016. mac Birmingham, a pioneering arts complex, now attracts over 1,000,000 visitors every year.

Birmingham's cultural infrastructure and its Creative Industries, like other cities globally, tend to cluster and form "cultural quarters" close to the city centre. This concentration of Creative Industries businesses has provided added value to the Greater Birmingham region, increasing visibility of the cultural product and encouraging the growth of the visitor and experience economies, as well as manufacturing and crafts.

Cultural organisations also act as anchors for development of strategies to promote diversity and inclusive growth. Birmingham Hippodrome has been active in promoting diversity through engagement with local communities to identify and develop the next generation of cultural and creative leaders. And the pressure on cultural organisations to identify new revenue streams, brought about by cuts in public funding, have resulted in innovative ways to identify and exploit repeatable IP.

Birmingham Cultural Investment Enquiry

The Enquiry is assessing the new, investment-based models of cultural investment and how they can be adopted in Birmingham. It sets out five key objectives:

- **Collaborative working in the CCIs** - recognising that the cultural sector's future is highly dependent on new and innovative forms of collective action. Examples include projects relating to skills development, big data and a new approach to city festivals are being developed under the Culture Central remit.
- **Public sector investment** – capitalising on renewed communications between ACE and Birmingham City Council, and the involvement of other city and sector partners, foster a more secure base for investment in the city.
- **Diversifying investment patterns** - in recent years the universities have developed important and innovative ways to support the cultural and creative ecosystem in line with their own institutional goals and with substantial benefits to the city as a whole.
- **More influential and commercially minded view to investment** – seek to establish a stronger line of responsibility and interconnectedness between the cultural sector and the business community, which is founded on an understanding of mutuality and joint success.
- **Creating a new city-wide cultural investment proposition** - BCC and others to establish ways to release the physical assets to the sector to help establish a new cycle of value creation. The sector to then work collaboratively to use this asset base to create additional value through equity, loans and endowments that provide new forms of income and financial sustainability. To engage with the anchor institutions in the city, in particular the HE sector, to create and enhance a new collaborative and progressive approach that recognises the multiple benefits that culture and creativity offer in highly cognate and strategic ways.

7.1.7 A centre of excellence for higher education

Greater Birmingham's Creative Economy benefits from the combination of higher education institutions across the West Midlands that provides an excellent supply of skills and talent from its young, digital and diverse population. A combination of world-class institutions, centres of excellence, first rate facilities and entrepreneurial spirit is creating a melting pot of hybrid digital industries. As the rate of technological change and innovation continues, 'tech' is becoming increasingly integrated within every sector and industry.

Birmingham City University (BCU) is the largest provider of graduates in creative disciplines of any university outside London and the South East. Its commitment to growing the Creative Industries sector and the wider Creative Economy is strongly reflected in Birmingham City University's STEAMhouse initiative. STEAMhouse is a creative innovation centre located within the creative community in Digbeth. It will demonstrate how arts and creativity can be combined with STEM subjects (Science, Technology, Engineering, and Maths) to drive innovation, skills, research, and economic growth, with the aim of brokering collaborative relationships between businesses, academics, arts and creative industries to support innovation and prototyping activity for social and economic benefit.

STEAMhouse is led by Birmingham City University in partnership with Eastside Projects, and funded by the European Regional Development Fund (ERDF), Arts Council England through the Creative Local Growth Fund and GBSLEP. It offers support to SMEs and new enterprises to develop new products and services through STEAMlab collaborative innovation workshops, grants, engagement events, and research collaborations with academics

Its reflects the importance of the wider Creative Economy, with themes around the growth sectors of low carbon, health and life sciences, advanced manufacturing, and creative and digital, as part of Greater Birmingham and Solihull LEP strategy.

STEAMhouse (phase 1) opens in February 2018 and runs until September 2019 – <http://www.bcu.ac.uk/business/steam/steamhouse>.

In addition:

- The West Midlands boasts three of the **top 20 UK Universities**¹²: Warwick (8th), Coventry (12th) and Birmingham (15th)
- Coventry is the top ranked UK University for **Film Production and Photography**¹³
- Warwick and Birmingham are the top two UK Universities for **Creative Writing**¹⁴
- Both Warwick and Birmingham are amongst the Top 20 Universities in the UK for **Computer Science**¹⁵
- Coventry is a top 10 University for **Journalism, Publishing and PR**¹⁶
- Birmingham City University¹⁷, Birmingham¹⁸ and Warwick¹⁹ are among the top 5 Universities for **Dance and Drama**

¹² Guardian University Rankings 2018

¹³ ibid

¹⁴ Complete University Guide 2017

¹⁵ Guardian and Complete University Guides, op cit

¹⁶ Guardian, op cit

¹⁷ Guardian op cit

¹⁸ Guardian op cit

¹⁹ Complete University Guide op cit

- The **Serious Games Institute**, a spin-out company of Coventry University, is a recognised world leader in the development of games education, virtual worlds and augmented reality.
- Computing centres of excellence include the **Computer Science Research Group (CSRG)** at Aston University and the **Centre of Excellence for Research in Computational Intelligence and Applications (CERCIA)** at the University of Birmingham.
- A £35 million Digital Plaza – **iCentrum** – is one of the first developments within **Innovation Birmingham**, a city centre Enterprise Zone on the campus of Aston University. The development sets the standard for fourth generation science parks and includes a software development playground and a tech incubation facility.
- The **New Technology Institute (NTI)**, part of Birmingham City University, produces graduates from super-vocational, industry backed postgraduate degree courses in video games development (**Gamer Camp**), digital marketing (**Future Media**) and film marketing and distribution (**Film Futures**).

7.1.8 Further education

The FE sector across Greater Birmingham is already working closely with Digital & Creative Industries employers both large and small. In the process, it has co-created some important and effective programmes for the sector; and has responded enthusiastically to the investments made by the BBC and the prospect of the relocation of Channel 4.

Birmingham Metropolitan College (Birmingham Met) is home to the only Digital & Creative Career College outside the M25. It is led by an Advisory Board which includes representatives from KPMG, BT, Glide (a local app development firm), social entrepreneur Joel Blake, social media visionary Jodie Cooke and James Cronin of Pakt Publishing, and Project Birmingham. The delivery team is made up of industry practitioners, working with students on live briefs from industry partners

to ensure that the content is tuned into current demand – such as a project brief from Amazon around social issues raised by digital engagement. As well as working on different types of brief - moving image, web sites, branding, corporate design, strategic presentations – students attend a weekly seminar where an employer comes into the college, or they go out to the employer's premises, to cover the kinds of 'soft' skills that will help them move into work, e.g.:

- How to use social media to raise your professional profile
- CV writing, interview skills, etiquette, presentation
- Financial planning – how to prepare for the 'gig economy'

In its first full year of operation, the Career College has engaged more than 20 employers and 400 students across a range of digital/tech (ICT, software, networking), digital creative (graphics, moving image, multimedia and photography) and management disciplines. The students that took part benefitted from connections and networking that have led them to set up microbusinesses, interactive websites, games and apps.

<http://careercolleges.org.uk/college/birmingham-metropolitan-college-bmet-creative-digital-technology-and-professional-services>

Quickcode Labs

Mick Westman, the former head of BT's Electrical and Engineering Academy and an industry veteran with 38 years of experience, worked with Solihull College, Solihull Chamber of Commerce and Innovation Birmingham to design and deliver a pre-apprenticeship programme for Level 3 students in digital and electrical engineering disciplines.

Adapted from models of 'learning by doing' – Westman is a Lego Serious Play practitioner – the programme provides learners with the core behavioural skills – attitude, aptitude and application - that employers are asking for. It uses blended learning – a combination of classroom teaching and projects set by local businesses, most of them SMEs – to deliver real outcomes in terms of web sites and business

process reengineering. As well as having to demonstrate application of technical ability, students are also shown how to build a business case for adoption of their solution.

Quickcode Labs is a social enterprise; projects are delivered at cost to local employers (or free where there is external funding available), making available resource that would be out of reach of most small businesses. Students spend half a day a week in the 'Lab' over the course of a 30-week programme, which is not tied to the academic year, and so provides further flexibility to meet employer needs.

The programme offers students the opportunity to progress to apprenticeships or other forms of tertiary education whilst learning 'on the job', and being paid – those that do go on to take undergraduate-level qualifications do so whilst building their professional portfolio with an employer, and without incurring large debts from student loans and fees, making it a more accessible route into highly skilled professional careers for individuals from a diverse range of socio-economic backgrounds. After a successful proof of concept with 20 students in 2016/2017, 100 students entered the programme in September 2017.

<https://www.quickcodelabs.com>

7.1.9 Sector intermediaries

Major cultural institutions such as Birmingham Hippodrome, Birmingham Rep and CBSO all have programmes to develop creative skills and cultural leadership capacity in Greater Birmingham. As we have seen above, universities and colleges are making significant investment not only to maintain a flow of skilled talent but also to help business leaders shape future commercial creative opportunities.

But smaller, grassroots organisations also make a major contribution to the cohesion of a cultural and creative cluster, the development of production skills and the creation and exploitation of new intellectual property.

7.2 Weaknesses of GBSLEP Creative Economy

Our statistical analysis revealed weaknesses in some sectors, geographies and skills, and in the lack of scale of individual firms.

7.2.1 Sector weaknesses

The following **sectors** saw falls in employment in Greater Birmingham and Solihull between 2010 and 2015:

- Film, TV, Video, Radio and Photography: -42%
- Publishing: -28%
- Museums, Galleries and Libraries: -12%
- Music, Visual and Performing Arts: -5%

All four sectors also saw falls in employment in both the Black Country and Coventry and Warwickshire over the same time period. Their economic importance has declined across the West Midlands – with reductions in public spending a contributory factor.

7.2.2 Lack of scale

There is a lack of **scale** in local creative enterprises:

- Crafts is the only sector in which more than 1% of enterprises in GBSLEP have 50+ employees.
- Over 5 years to 2015, the number of companies with 50 employees or more declined in advertising and marketing, architecture, and IT.
- The creative industries in Greater Birmingham and Solihull are dominated by microbusinesses (0 to 9 workers) – with all the dynamism and vulnerabilities entailed with operations of this scale.
- To balance this, there is some evidence of strengthening scale via transition from sole proprietor businesses to microenterprises.

Within the DCMS sectors, 8% of creative businesses in GBSLEP in 2015 were sole proprietors, compared to 17% in 2010; while 86% of creative businesses in GBSLEP in 2015 were microenterprises, compared to 76% in 2010.

7.2.3 Physical infrastructure constraints

We carried out qualitative research through a series of workshops. This identified some other weaknesses including:

- Difficulty in finding affordable workspace
- Birmingham's road network hinders, rather than aids, ease of movement around the city (particularly for pedestrians) and presents barriers to growth of clusters
- Other transport links across the city region are inadequate and poorly coordinated, with a direct effect on people's ability to attend cultural events and access employment opportunities in other districts.

"The development of more affordable office spaces is highly desirable for future growth of the sector, and should be part of broader regional growth strategies ... driven by new and existing devolved authorities, and supported by the LEPs."

Source: The Association for UK Interactive Entertainment (UKIE) (2016) "The UK Video Games Sector. A Blueprint for Growth"

7.2.4 Digital infrastructure

- Digital connectivity for businesses in Greater Birmingham and Solihull lags behind that of other cities. Some 20,000 business premises in the LEP area – 28.7 percent of the total – have average download speeds of less than 30Mbps.
- Only 41 percent of business premises in the GBSLEP area are served by broadband connections capable of average download speeds of 100Mbps or more.²⁰
- These figures relate to modelling of ‘consumer’ broadband supply – in reality, many fewer creative SMEs will subscribe to business grade services offering symmetrical connections of 100Mbps up to 10Gbps, needed to facilitate transfer of large files and management of distributed teams essential to the growth of sectors such as games and VR.

7.2.5 Lack of reputation as a creative city

- “Birmingham” is not a city brand that is associated with creativity in the same way as comparably sized cities – Manchester, Berlin, Barcelona – even though Birmingham outperforms Manchester on some indicators of creative strength!
- Lack of larger businesses in the sector means that there are fewer creative leaders with the authority to act as sector champions, and fewer brands that are emblematic of Greater Birmingham.

7.2.6 Visibility of cultural and creative industries

- Expectation (particular amongst cultural organisations in Birmingham) that a paternalist City Council will ‘do the right thing’ leads to a culture of dependency.
- Cultural organisations need to assert their presence – and local authorities (particularly Birmingham City Council) need to give them the physical and digital ‘real estate’ to express their contribution

7.2.7 Silo Working

Geographic Silos

- Most clustering is within the core city area, rather than across the wider geography of GBSLEP.
- Some areas (Tamworth/Wyre Forest) are particularly lacking in clusters.

Sector Silos

- Siloed working practices within creative sectors limit transfer of skills
- Gaps exist between commercial creative and community/ cultural sectors – no shared sense of an overarching creative community.
- Not enough lateral thinking or evidence of collaboration between the arts and other economic sectors, or wider society

7.2.8 Access to Finance

- Evidence from Innovate UK and Creative England suggests that, as a region, creative businesses in the West Midlands fare worse than those in comparator geographies in terms of the number and success rate of applications for public funding for project development and R&D.

²⁰ Point Topic, December 2016

- Private sector finance (in the form of angel and VC investment) is not distributed equitably
- SMEs need help to 'learn the language' of financial applications
- Cultural organisations need access to more revenue funding

7.2.9 Skills

- Fewer Level 4 workers in GBSLEP than comparator city regions

- Across the UK, at 36% of creative industry businesses, 80% of workers are qualified to Level 4 or above, whereas only 25% of creative industry businesses in the West Midlands are qualified to this level. (UKCES)
- Need for supply of skills to keep pace with rate of growth in employment in those sectors that are developing most rapidly

7.3 Threats to GBSLEP Creative Economy

7.3.1 Continuing austerity

- Austerity and cuts in public expenditure since 2010 can be seen to have hit arts and cultural organisations both large and small – employment in those sectors continues to fall.
- Continuing wage restraint in public sector has a knock-on effect in reduced footfall in the city centre, affecting demand for small craft producers, specialist cinema, live music etc.
- Public sector cuts have driven the biggest falls in creative sector employment and further cuts would continue this trend.

7.3.2 Brexit

- Loss of freedom of movement and uncertainties around rights of EU citizens to remain in the UK may result in an exodus of skilled programmers.
- The direct effect on GBS employers may be multiplied by a 'brain drain' as larger companies in London and the South East offer bigger salaries to lure talent from other parts of the UK.
- This could have a 'ratchet' effect, where employers are forced to increase salaries to seek to retain the remaining pool of people with relevant skills, increasing costs and reducing competitiveness. It may lead to creative businesses having to turn down work due to their inability to access suitably skilled

team members – a factor already observed in other UK core cities, such as Manchester.²¹

7.3.3 Falling demand

- Advertising, marketing and digital agencies may see a fall in demand as other companies reduce expenditure in an attempt to mitigate the effects of reduced demand and price inflation.
- Consolidation of independent TV production companies continues; should local independent producers fail to secure commissions to replace the demand represented by 'hit' shows (such as Maverick TV's 'Embarrassing Bodies'), employment in audio-visual sectors may continue to fall.

7.3.4 Displacement of existing businesses

- Strong performance in crafts is very much associated with the Jewellery Quarter – an area much in demand for residential developments. Policy and planning need to reconcile traditional strengths of Jewellery Quarter with demand for more recent residential and leisure development.
- Displacement by digital technology of jobs in sectors that have already suffered falls in employment in Birmingham – such as publishing, TV – where digital technology means that some jobs are less labour-intensive

²¹ Manchester Digital Skills Audit Report (2016) - <https://www.manchesterdigital.com/sites/default/files/Skills%20Audit%20Report%202016.pdf>

7.3.5 Sector may lack resilience

- With such a large proportion of micro-businesses, and evidence that creative businesses in Greater Birmingham & Solihull are on average reducing in size, will the sector be more vulnerable to economic downturn?

7.3.6 Underinvestment in digital infrastructure

- We have already noted the lack of access to high speed broadband compared to other core cities. The threat of dated and inadequate digital infrastructure becomes more acute as digital production and consumption of creative services moves increasingly online.

7.4 Opportunities for GBSLEP Creative Economy

7.4.1 Inward investment

Recent inward investment successes, particularly in IT (e.g., Advance, ASOS), and the arrival of BBC THREE, not only create employment opportunities but increase supply chain opportunities. They also reinforce the status of Greater Birmingham and Solihull as a creative hub and create a 'halo effect' for further investment. The sense of purpose and confidence leant by these recent successes was evident in the combined West Midlands response to DCMS consultation on the relocation of Channel 4; it is important to retain momentum.

7.4.2 Indigenous growth

The creative industries across Greater Birmingham have a number of potential 'poles' of growth, with evidence of clustering, both geographically and in sectors that demonstrate local strengths.

There are also positive indicators of recovery in sectors that have experienced a downturn. The continuing growth of in-house and independent production from BBC Studios and growth of other

independent TV production companies combine to arrest, and then reverse, decline in television and film employment.

The cultural sector can also benefit from local examples of organisations that have successfully diversified their sources of income away from 'at risk' public funding through developing and exploiting IP in local and international markets; growing markets for their skills and IP outside the cultural sector; and engaging with a new generation of philanthropists more interested in the social outcomes of their work.

7.4.3 Build social capital to achieve inclusive growth

The diversity of the people and communities of Greater Birmingham and Solihull is now routinely identified – rightly - as a strength and a source of potential. But evidence from across the creative industries reveals ongoing, and in some cases worsening, challenges around diversity.²² It is essential the industry improves the recruitment, retention and advancement of women, BAME, disabled people and those from less advantaged backgrounds to remain economically and creatively competitive.

7.4.4 Productivity growth

The role of technology in transforming skills and productivity in other areas of creative services and manufacturing, such as advertising, computer services and design, is reflected in the way in which these sectors are growing in step with each other; and in their predominance in both employment and output across the creative economy of the GBS LEP area.

²² Work Foundation (2017). "A Skills Audit of the UK Film and Screen Industries. Report for the British Film Institute." http://www.theworkfoundation.com/wp-content/uploads/2016/10/420_A-Skills-Audit-of-the-UK-Film-and-Screen-Industries.pdf

The evidence of people in creative occupations working outside those sectors points to an increasing exchange of people, skills and ideas between sectors such as IT, Design, Advertising & Marketing and other high value add sectors in the region (business and professional services, advanced manufacturing). Indeed, GBS LEP's own definition of Business, Financial and Professional Services includes Advertising and Marketing. This interaction is an increasingly valuable source of commercial opportunity and collaborative development of new products and services. The scale of employment opportunities in these sectors is reflected in the large number of job vacancies on LinkedIn – for example, over 900 marketing jobs advertised in Birmingham alone. This means that both companies and individuals looking to relocate to Greater Birmingham can do so in confidence, the employer knowing that there is both a pool of available labour and the individual knowing that there will be opportunities to progress to second, third and further jobs in the wider West Midlands region if they opt to take a post in the region.

"Design has a role to play in engaging SMEs and communities in local economic development. Design can translate science and research into valuable assets for business, people and places, encouraging its uptake."

Design Council (2017). "Design: Delivering a New Approach to Growth"

Interdisciplinary initiatives such as STEAMhouse should further strengthen the Design cluster in Birmingham, and increasing awareness of the value of other regional specialisms (in data analysis and behavioural insight) has potential not only to drive innovation but to increase productivity across all sectors of Greater Birmingham's economy.

7.4.5 Planning and infrastructure

The 'overspill' of software, design and 'making' from the city centre in the form of increased clustering across the West Midlands will be assisted by improved connectivity, including the metro extension.

But it is important that development of creative quarters should recognise existing clustering and use devolved powers to minimise displacement of cultural and creative organisations:

"England has this year seen the election of new metro mayors [in six city regions, including the West Midlands], and devolution deals with combined authorities are unlocking potential for greater economic autonomy across the country. The next step should be to support a further transfer of powers on housing and planning from Westminster to our local communities and regions to help shape a better built environment. A planning system that is supported by local people will only be possible with higher levels of public trust. Section 106 and Community Infrastructure Levy (CIL) agreements should not be subject to commercial confidentiality, which allows some developers to weaken their obligations to deliver benefit to communities from new developments."

RIBA Manifesto (2017). Building Global Britain

7.4.6 Placemaking and promotion

Continuing regeneration of areas such as Smithfield in Birmingham and big ambitions from Combined Authority on house building will contribute to a wider offer on 'quality of life', as well as availability of workspace.

Opportunities exist across the region to change perception of the West Midlands, and Birmingham at its heart, as a cultural and creative hotbed. The success of Coventry's bid to become Capital of Culture 2021, and other bids in progress, including Birmingham's bid to host the Commonwealth Games in 2022 and West Midlands Combined Authority's proposal to relocate Channel 4 to the region, could,

individually and collectively, add to the profile of Birmingham and the West Midlands as the cultural 'heart' of the UK. All of these proposals would also increase demand for housing, services and cultural provision in a way that would boost the economy of the GBSLEP area – for example, the regional impact of Channel 4 relocation is estimated at £5 billion over ten years.

8. Analysis of findings

8.1 What are the main trends affecting the sector?

- Increasing reliance on data and data analysis
- AI and machine learning – not taking away jobs, as the tabloids would suggest, but shifting focus of value creation – where the benefits of data analysis and behavioural insight by the machines can be translated by creative individuals and teams into new, and higher value, content and services.
- Augmented and virtual reality

8.2 How are these manifesting themselves?

- Digital disruption of **production**: e.g., data analysis and AI dramatically changing patterns of work in advertising; digital manufacturing ('fab labs') transforming design, prototyping and production processes, with the largest effect being felt in sectors other than creative and digital but making agencies and individuals with those skills an indispensable part of the supply chain for advanced manufacturing, etc
- Digital disruption of **consumption**: consumers of entertainment content moving relentlessly online, with fragmentation of audiences across platforms. Although often accompanied by growth in overall audiences, online revenues do not yet replace those in 'traditional' media – sectors such as music, journalism, publishing having to find new business models to adjust to those changes
- Much greater interest in '**immersive tech**' from government and private investors, both for entertainment, other high growth verticals (adv manufacturing, logistics, health) and visitor economy

8.3 What is the impact of Brexit?

- Increased costs (both tariff and non-tariff barriers) in distribution of entertainment content, both in physical and digital form
- Uncertainty about future regulatory requirements, particularly around personal data and privacy
- Increased cost and complexity for live performance (theatre, music) touring in the EU – visa, tariff barriers and additional administrative time and expense
- Risk that EU workers will leave, exacerbating skills gaps and inflating wages for key skills (especially digital)
- Inability to recruit EU workers, exacerbating skills shortages
- For the cultural sector, inability to get work permits for EU performers

8.4 Which sub-sectors are most significant in GBSLEP and the West Midlands?

- IT and software: 14,000 jobs and £1.2 billion GVA in GBS LEP (2016) and a further 8,000 jobs in CWLEP
- Advertising and marketing: £327m GVA in GBSLEP
- Although smaller, design and craft sectors also remain important [are growing?], are developing clusters in [our local towns] and have influence across the economy – 57% of design and crafts employment and GVA outside CI
- Nationally significant crafts cluster in Birmingham (and Lichfield)

- Although the account of its recent performance is mixed, the **high-end content production sector** remains a strategic priority (young, diverse, digital, important for representation and aspiration of citizens, raising profile of Birmingham as a cultural catalyst, new business models). This sector has reached the ‘inflection point’. The official statistics end at a point where the independent tv and film sector was just starting to witness a rebirth – here’s the qualitative evidence for that – and we are seeing both a resurgence in confidence, new investors (BBC THREE), skills (BBC Academy) and plans for big capital investments that together have the potential to re-establish Birmingham and the surrounding city region as one of the major centres of content production outside London.
- how contraction can be reversed, including existing initiatives and future investment proposals - and map where the GBS and other actors will combine to deliver a strategy for growth including some or all of: WM Film Fund, Mercian Studios, production village, lobbying for higher and more tightly enforced clusters for out of London production.
- For the West Midlands as a whole: video games (c 3,500 jobs and £250m GVA in 130 companies, focused on Leamington Spa which has 75% of that employment and higher productivity)

8.5 Which locations within the LEP area are significant (and showing fastest growth)?

- Birmingham City Centre – IT/software, advertising and marketing, design and crafts are all strong.
- Most of the region’s film and television production is focused on Birmingham, but there is increasing evidence of activity in Solihull and proposals for investment in Mercian Studios and

a production village would increase this greatly, rather than displacing activity from the city centre

- Solihull, Bromsgrove and East Staffs – evidence of clustering in IT/software

8.6 Where are the main competitors (for investment)?

- London – 75%+ of all creative FDI projects never look at any other UK city
- Greater Manchester – both MediaCityUK (TV and digital) and Manchester City (digital agencies, fintech)
- Bristol – IT/software, animation, TV/film, AR/VR
- Edinburgh – AI, AR/VR, machine learning, fintech

8.7 What are the key advantages of a West Midlands location for creative and digital businesses?

- Accessibility – transport links
- Cost of premises – though city centre supply tightening and prices rising, in part due to competition with space for residential development
- Housing supply, type and cost
- University specialisms in data analysis, AI and HCI (human computer interaction, via interface design, games etc)

8.8 What are the key challenges to the sector in GBS LEP?

- Image: to date, Birmingham and the West Midlands has not been seen as a creative region on a par with Manchester/Liverpool or Bristol
- Linkages within the creative/tech ecosystem are poorly understood (even by businesses themselves, in many cases)
- Self-confidence: the ‘shy Brummie’ syndrome
- Inclusive growth: promoting the benefits of sector growth to the region’s famously young and diverse population

- Skills levels: a deficit in NVQ4+ skills compared with national average and main core city competitors
- At present Birmingham and the West Midlands arguably has cost advantages over competitors - London, Manchester, Bristol and Edinburgh – but no clear proposition as to why this is the best place to access new market opportunities

9. Conclusions

GBSLEP has a huge and dynamic Creative Economy employing 49,600 people and generating £4.1 billion GVA per annum.

- 45% of Creative Economy employment and 49% of GVA is from outside the Creative Industries. This reflects a core of digital skills, both in IT, Software & Computer Services and Advertising & Marketing, that is critical to the competitiveness and growth of the Creative Industries and the wider economy.
- Advertising & Marketing (£1.6 billion GVA, 14,350 jobs) and IT, Software & Computer Services (£1.5 billion GVA, 17,900 jobs) jointly contribute 75% of Creative Economy GVA and 65% of Creative Industries employment.
- The dispersal of advertising and marketing skills across the economy - 80% of those working in advertising and marketing work in businesses outside the creative industries - is reflected in the very large Business, Financial and Professional Services (BFPS) sector of GBS LEP
- Digital content, services and manufacturing together represent a nationally significant cluster across Greater Birmingham & Solihull and Coventry & Warwickshire.
- Clustering of craft skills and production in and around Birmingham is strongest of any of the LEPs.
- Although concentrated in Birmingham city centre, the creative economy has emerging strengths (in particular, in Advertising & Marketing, IT, Design and Crafts) across most areas of the conurbation.

Despite this encouraging evidence, our overall assessment is that **the potential of the Cultural and Creative Industries in Greater Birmingham remains unfulfilled.**

How do we reach this conclusion? In analysing the comparative performance of the Creative Industries in Greater Birmingham against comparator LEP areas, we see:

- Creative firms represent a lower proportion of the business base and are less strongly concentrated within Greater Birmingham than in other knowledge-intensive clusters around Oxford, Cambridge and Bristol
- Greater Birmingham has lower levels of creative employment than other core cities such as Manchester and Leeds
- Looking across Greater Birmingham, the opportunity for further, inclusive growth through Creative Industries has not been realised in localities such as Tamworth and Wyre Forest.
- Recent discussions around the potential relocation of Channel 4 highlight that the high-end production sector is at an inflection point. Employment in some core content creation sectors has contracted over the period 2010-2015, although it still supports 22,000 jobs in the creative economy. There is evidence of new investment, including that by the BBC, but television, film, games and digital content production all require sustained focus and continuing investment to ensure their continuing competitiveness.
- Employment in 'core' artistic employment – Music, Performing & Visual arts -showed a decline in the city core over 2010-2015. This parallels a similar fall in Museums, Libraries & Galleries, influenced by falling levels of public spending and displacement pressures brought on by increasing cost and redevelopment of areas of the city centre. Other towns in the GBS LEP area are seeing some increases as artists extend their search for affordable studios; but there needs to be a strategy to address the risk of displacement in this sector and to find new business models to replace an over-dependency on public funding.

10. Recommendations

We conclude with recommendations to GBSLEP for the growth and development of the creative economy within Greater Birmingham & Solihull.

The success of the creative economy within Greater Birmingham is critical to achieving targets for growth within the West Midlands' economic strategy – and reflects priorities of national industrial strategy to identify place-based clusters, linked to strong institutional structures equipped with the appropriate resource to accelerate local growth.

To address this, steps need to be taken to address:

- **Productivity and skills**, building on strengths in key sectors, mitigating risks of further decline in weaker sectors and addressing the need to build social capital to promote inclusive growth by increasing the diversity of the cultural and creative industries.
- **Infrastructure**, including workspace, planning of new developments, transport and digital connectivity.
- The **governance and culture** of the cluster, including convening sector structures across the West Midlands to address confidence and cohesion of the sector and the educational and industrial strategy needed to underpin inclusive growth and additional investment.

10.1 Productivity and skills

10.1.1 Quickly move to articulate the creative cluster offer

GBS LEP and city regional policy needs to articulate the offer of the creative cluster, as well as the requirement for intervention. National

policy for the creative industries - in particular the Industrial Strategy White paper published in November 2017 - emphasises the need to go beyond business support to identify a place-based strategy that recognises the appropriate local mix of employers and institutions to drive it forward.

Although specific funding has not yet been identified, GBS LEP and West Midlands Combined Authority policy needs to prioritise and define specific offers, eg:

- Addressing evidence of barriers to scaling through a programme of events, training and resources to help creative entrepreneurs to grow their businesses and get investor ready.
- Bringing investors and local creative businesses together.
- Promoting local heritage and culture, including crafts, design and high-end content production, as a source of regional competitive advantage.
- Underpinning the supply of talent through a range of industry-led approaches to skills in schools, colleges and universities.
- A digital skills strand for the wider Digital & Creative Industries sector, emphasising the importance of data analytics and behavioural insight in defining new products and services and creating new IP.
- Extending the games sector specialist role, developed by Coventry & Warwickshire LEP in partnership with Ukie, to cover the whole West Midlands region.

Inclusive growth means spreading opportunity geographically, as well as in terms of demography and sectors. There are 'hot spots' of Creative Industries activity and clustering in Bromsgrove, Lichfield and East Staffordshire, as well as Birmingham and Solihull; but Redditch, Tamworth and Wyre Forest lag behind.

10.1.2 Diversity and building social capital

The city region boasts of its diversity – but it isn't yet adequately reflected in the creative sector. Aside from putting an unnecessary and artificial limit on supply of skills, this impedes both development of cultural leadership in BAME communities and the dispersal of businesses and employment opportunities across the city region.

Diversity is a cross-cutting consideration: increasing social equity is a priority to be addressed in all cultural and economic development policy for the sector. If a proposed intervention cannot demonstrate how it will work for the population of the whole city region, it shouldn't receive public support. Any strategy to promote diversity must give people real engagement in and control over decision making, not just 'ticking the box' – and should reap economic benefits in terms of increased awareness and exposure to diaspora markets, other kinds of cultural and economic experience.

Local projects such as RE:Present and ASTONish have demonstrated that engagement with culture helps build social capital. Opportunities to experience of leadership in the cultural sector translates into greater willingness amongst minority groups to take advantage of entrepreneurial opportunities, not limited to culture and the creative industries.

This is reflected in the importance given to audience development by national institutions, such as the British Film Institute. This recognises that film can play a role in, 'drawing communities together through their mutual enjoyment of cultural experiences; as a means of enriching a sense of local identity through locally produced content or content

featuring local subjects or interests; and as a catalyst for social action across other agendas locally and regionally.' ²³

10.1.3 Accelerate growth in sectors that display existing strengths

Data analysis and behavioural insight

- AI and machine learning – not taking away jobs, as the tabloids would suggest, but shifting focus of value creation – where the benefits of data analysis and behavioural insight by the machines can be translated by creative individuals and teams into new, and higher value, content and services.
- Augmented and virtual reality

Future proofing the sector and employment opportunities, in particular for the young and diverse population looking to the sector to provide careers and livelihoods long into the future.

GBSLEP already has some exemplary programmes that seek to link investment in skills to industry needs. But the city region and its institutions could go further in developing programmes that build resilience amongst the leaders and managers of cultural organisations and creative businesses to ensure that they are equipped to grow successful businesses and continue to invest in new IP.

10.1.4 Exploit culture and heritage

A powerful cultural offer means that there is capacity and infrastructure available, particularly within the city core, that can be harnessed for the development of leadership skills, diversity and more inclusive growth.

²³ DCMS (2012). "A Future For British Film. It begins with the audience..." (UK Film Policy Review) http://old.culture.gov.uk/images/publications/DCMS_film_policy_review_report-2012_update.pdf

Related to this, the craft heritage of Birmingham, the ‘city of 1,000 trades’, still has much to offer, and not just in terms of the visitor economy. KPMG’s report for the Crafts Council (2017) sets out a recommendation for LEPS that we would endorse here:

“Facilitating the collaboration between businesses and other sectors at a local level, potentially through Local Enterprise Partnerships, local authorities and growth hubs, with support from the Crafts Council to help to ensure business support, training, advice and mentoring is fit for purpose.”

Craft skills and knowledge have a strong economic impact and significant potential to drive further growth and innovation in other sectors, as this report demonstrates.’ There are, though, barriers that stand in the way of realising that full potential. These fall into three broad areas: lack of understanding of the value of craft innovation, positive externalities and a degree of risk that lead to underinvestment in innovation by individual firms, and underinvestment in craft education and skills.”

Source: KPMG (2017). "Innovation through craft: Opportunities for growth. A report for the Crafts Council"

10.1.5 Recognise that innovation comes bottom up as well as top down

Some of the most innovative, internationalised and successful IP-based business models in the creative industries come from arts and cultural organisations. In Greater Birmingham, this includes Stan’s Café and Ikon Gallery

10.1.6 Reconstruct a high end creative content cluster

Whether Channel 4 relocates to the West Midlands or not, the alignment of the content production sector (including games) to a 21st century, digital way of doing things makes it essential that forward-looking content businesses should have a foothold here.

10.1.7 Make use of the WMCA proposition in key growth sectors

In areas where inward investment is key to continuing to grow competitiveness and productivity, such as the high-end production cluster and IT (and, in particular, games), there is a compelling argument for the business case to reflect the capacity and capabilities of the combined West Midlands region. This can help turn good propositions into great ones; builds on the excellence of the knowledge base in areas such as data analysis and behavioural insight, where the Universities of Warwick and Coventry have particular strengths to add to those in GBSLEP; and, in what is bound to become an increasingly ‘tight’ market for FDI, ensures that the region can win the right to be the second place for FDI in creative and digital sectors after London.

10.1.8 Scale

Although the overall picture is one of growth, individual creative businesses in GBSLEP seem to have a particular difficulty in scaling up – in fact, average company size in the sector is falling. We have yet to fully analyse the causes of this but, whilst some of the structural failings are common to other core cities, the repeated anecdotal evidence of ‘shy Brummies’ may be creating other, self-imposed barriers to growth.

However, there seem to be fewer problems associated with scaling in the creative economy. We need to look at examples in other sectors to understand how the scale and structure of the creative economy in GBS and the West Midlands region provides impetus for skills, employment, productivity growth and inward investment. Within that, where are the strategic market opportunities for individual creative sub-sectors? Where should we place the focus?

10.1.9 Address structural shortcomings of market for risk capital outside London

Structural failings of Venture Capital for IP-based businesses outside London are recognised. Lack of data about opportunities and returns from previous investments leads to fewer people from within the sector taking the decision to reinvest their profits in their own R&D and in other early-stage ventures, and fewer experienced angel investors from other backgrounds being attracted to the sector. This results in businesses accepting too little investment at each 'raise' to effectively mobilise R&D and commercialisation of original IP – so the UK creates the innovation, but overseas owners reap the commercial benefit.

Public sector (e.g., MEIF) and private investors (e.g., Advanced Digital Ventures, which has backing from British Business Bank as well as private investors) are striving to address those barriers, but to do so requires granular information about angel networks and existing deals that don't appear on databases such as Crunchbase. This is one area in which the cultural and creative sector could help itself by coming together to address the information gaps so that the public sector and private finance markets can more effectively address opportunities to develop and exploit innovation across the sector.

10.1.10 Make skills fit for purpose

This is one of the key areas in which the combined West Midlands offer is world class – the mix of research and applied disciplines from Birmingham, BCU, Warwick and Coventry Universities in areas of data analysis, behavioural insight and disruptive media, with applications in

Computer Games, Advertising & Marketing and EdTech. Its excellence needs to be understood and promoted to investors and businesses in other sectors.

- Birmingham has a good precedent in harnessing the experience of sector specialist intermediaries to work with employers and providers to increase matching and match-fitness of new entrants to opportunities. This needs to be embedded in the GBSLEP approach to skills and explored at a West Midlands level, to take advantage of diversity and youth of city region's population.
- Partners across WMCA need to take steps to increase the number of people in the city region qualified to Level 4 and above in relevant creative, scientific and technical disciplines, including design, maths, physics, data analysis, behavioural insight and applied creative and audio-visual production disciplines.
- West Midlands universities need to strengthening links with business to encourage more graduates to feel that they can build careers, or businesses, in Greater Birmingham and the wider region, taking advantage of lower costs, great connectivity and a higher quality of life.
- FE and HE institutions across Greater Birmingham and Solihull, and academies like Birmingham Ormiston Academy, are working closely with Creative and Digital employers to tailor their programmes to help both address short term skills shortages and to close the gap in the number of people with Level 4+ skills. This is an area in which the scale of the offering to employers and investors is assessed and amplified across the geography of the city region, for the benefit of locations across the West Midlands.
- GBSLEP can help promote awareness of opportunities in the creative industries to learners, parents and employers by highlighting this investment and by championing the best practice examples of skills and apprenticeships, such as Creative Alliance's 'Multi-Employer Model' and 'Micro-Module Model'.

10.2 Infrastructure

10.2.1 Integrated approach to address 'quality of life' and attract inward investment

A city region-wide integrated strategy for transport, housing, culture and liveability would help to:

- Harness the 'diversity dividend' for all citizens and businesses.
- Highlight the value of the city region's cultural offer in increasing wealth and wellbeing for the whole population, increasing its profile and status in policy and planning.
- Increase awareness of opportunities for employment and development for all.
- Improve graduate retention.
- Attract new inward investment.

10.2.2 Put strategy and planning policies in place to deal with displacement before it undermines key growth sectors

All cities have a problem in identifying enough affordable workspace, particularly for those working in Creative Originals and Creative Experiences. At the moment, Greater Birmingham seems not to have an acute problem above that of other core cities – but without the right approach to planning of new developments, particularly around HS2 in Birmingham and Solihull, and existing creative clusters such as Jewellery Quarter and Digbeth, it could become chronic. Early intervention to establish policies to earmark Section 106 and CIL monies to specific cultural and creative uses, for example for investment in affordable workspace or cultural provision, and to enforce Article 4 provisions to limit conversion of commercial space to residential, could help retain and build upon the cultural grain of Birmingham City Centre and help accelerate growth of creative clusters in other towns. GBSLEP can support this by encouraging local authorities to:

- Encourage local authorities to adopt of the 'agent of change' principle to safeguard performance venues against threat of displacement, by ensuring that planning conditions for any nearby residential development cannot threaten the licensing of the pre-existing venue.
- Ensure that plans for a creative quarter at UK Central within NEC, in Solihull, reflect the need to transform project-based investment in 'out of London' production of television and film into permanent employment bases.

10.2.3 West Midlands Digital Transformation plan

- The approach taken by GBSLEP to mapping its Creative Economy, including the economic impact of creative occupations in other sectors, should be adopted and extended by the West Midlands Digital Transformation plan to include mapping of the whole 'digital ecosystem' – from infrastructure at the level of international connections, internet exchanges and data centres, through assessment of the economic impact of cross-sectoral supply chains for the Creative Industries and digital manufacturing, to mapping angel networks and VC investment, and all points in between.
- The West Midlands needs a way to measure and promote its pool of digital talent. It should consider adopting approaches such as Manchester Digital's annual skills survey as a way to help employers, investors, students and those considering the opportunity of re-skilling understand the skills shortages, gaps and opportunities in digital employment across the city region.
- WMCA has submitted an Expression of Interest to DCMS's Local Full Fibre Network programme, and expects to follow up with a full funding bid to 'Wave 2' of the fund in January 2018. This needs to be more ambitious than the 'Wave 1' voucher scheme pilot in Coventry & Warwickshire and focus on building the infrastructure needed to reflect and serve the increasing

demands of producers and consumers for digital connectivity, both fixed and mobile.

10.2.4 Build the night time economy

One way to both boost consumption of Cultural and Creative Industries and to promote development of supply chains in sectors such as Music and Performing Arts is to support an integrated planning and licensing framework to support the development of the night time economy.

This would have additional benefits in changing the perception of Birmingham to one of a cultural hot-spot in the eyes of audiences, investors, students and potential workforce.

“The Mayor of London has successfully established a Music Development Board and appointed a Night Czar to maximise the potential for the capitals night time economy. Similar measures should be considered by other elected Mayors and considered as part of devolution deals.”

Source: UK Music (2017). UK Music Manifesto

10.3 Governance and culture of the cluster

10.3.1 Sector leadership

Creative Industries need a stronger voice in local and regional decision making. The formation of the West Midlands Combined Authority (WMCA) supports the ambitions of Greater Birmingham for creative businesses: WMCA is aiming to grow the Creative Economy of the city region by £7bn and 29,000 jobs, by 2030. And the election of a WMCA Mayor has sharpened the focus on the potential for the holder of this post to act as ‘convenor’ for the sector.

The LEP should support development (at a Combined Authority level) of:

- The role of the city region mayor as convenor of creative and digital transformation.
- Industry-led structures to advise and scrutinise quality and coordination of interventions by public sector and institutional actors.
- The importance of recognising and measuring the economic, social and cultural value of very different kinds of cultural engagement – from Birmingham Mela, the UK’s largest Asian Music Festival, with an annual audience of 70,000, to steps taken by BFI to increase audience development for film and moving image to the kinds of mapping of informal cultural activity (dance, eating out) and social value undertaken by Prof Jonothan Neelands at Warwick Business School as part of Coventry City Council’s cultural planning.

The ‘WMCA Culture and Creative Sector Overview’ presented by TFCC in September 2016 makes a point about comparing the organisation of the culture sector in WMCA with that in other city regions, citing examples from Liverpool and the Tees Valley, both of which reference culture in their SEPs.

- This was borne out in the arts and culture workshop that took place as part of the GBSLEP Creative Economy Mapping, in which participants recognised the benefits of a partnership structure similar to that of the Liverpool Arts Regeneration Consortium (LARC) in addressing policy and dissemination.
- The ‘Creative Fuse North East’ project provides a ‘live’ example of a city region-wide approach that builds on an existing cultural partnership between arts, business, local authorities and Universities to focus on the benefits of merging creative and digital capacity and capabilities in a regional approach to cultural and commercial development.

All these points are recognised in the **Birmingham Cultural Investment Enquiry** – and we would reflect TFCC’s recommendation, ‘*to roll out the learnings from the enquiry*’ whilst emphasising that there are already structures in place – notably, Culture Central – that could form the basis of further development in the sector in order to enhance cohesion and coordination and minimise the risk of unnecessarily duplicating structures and initiatives across the West Midlands.

10.3.2 Raise confidence in and awareness of GBSLEP as a creative hub

Arts organisations and industry professionals across GBSLEP report a lack of confidence in putting themselves forward. This seems odd, particularly in an environment where the city region’s Digital Industries, Design and Advertising & Marketing sectors are significant and growing contributors to employment and wealth, and arts organisations have themselves shown demonstrable success in diversifying sources of revenue and funding.

Starting with members of the cluster themselves, the Cultural and Creative Industries need to:

- Free up cultural organisations from city patronage and a mindset of ‘asking permission’ rather than ‘begging forgiveness’ - an area

in which, at a Combined Authority level, the convening power of the Mayor could have great impact in opening up decision making at different tiers of local government.

- Recognise that great IP-based models in the cultural sector as well as in industry – eg Ikon, which is better known in international markets than in Birmingham for its mix of curation of original touring exhibitions and publications based on those shows; and Stan’s Café, whose ‘Of All Of the People In All Of The World’ (known as ‘the rice show’) is an innovative example of a scalable, low-cost, theatrical format with huge international licensing potential – and advocate that other arts organisations and creative businesses investigate their use as a way to identify their own distinctive approaches.
- Shout about how great Greater Birmingham is as a place to live and work, and emphasise that its residents enjoy vastly higher monthly disposable income than Londoners.

10.3.3 Build on strength of cultural ‘anchor institutions’

The evidence of success of arts and cultural organisations is reflected in the recent announcement of increased Arts Council England funding (to £88 million per annum) and an increase in number of National Portfolio Organisations (to 44). But the pressures on other parts of the cultural sector, especially those that demonstrate high levels of dependence on public funding, remains.

A mix of actions at local authority, LEP and Combined Authority level is needed to address sustainability of key sectors of commercial and cultural activity.

- For Museums, Galleries & Libraries, GBSLEP and its constituent local authorities should consider whether alternative models (eg, cooperatives and trusts) might offer greater resilience than public ownership in preserving and extending the use of public cultural services and assets.

- The same partners need to come together to address the structural weaknesses of arts organisations and cultural institutions that are, individually and collectively, overdependent on public funding in a time of austerity – through diversifying the funding base, in ways aligned with ACE Catalyst and similar initiatives, and by providing expert support for innovation in business models and exploitation of IP.

regional offer for film and television that amplifies the regional strengths of the Creative Content sector and makes a strong case for investment.

10.3.4 Take steps to mitigate risk of further decline in weaker sectors

The fall in employment in some Creative Industries sectors observed in our mapping reflects both structural weaknesses and the impact of external factors such as commercial divestment, digital disruption of markets, displacement resulting from commercial development of space and the effects of ‘austerity’ in reducing public funding. Once ‘critical mass’ in these activities is lost, it is much harder to recover.

10.3.5 Recognise and empower appropriate intermediaries

Universities and colleges across Greater Birmingham have taken the initiative in engaging employers across the Digital & Creative Industries; but plans to support and grow the sector should also recognise and utilise the experience and expertise of a range of local festivals, arts organisations and community-led activity in delivering appropriate and effective interventions to improve rates of start-up, ‘stay-up and scale-up’ amongst creative entrepreneurs.

10.3.6 Work regionally to deliver investment and growth

This proposition is all about scalability, so is clearly more transformative when delivered on a regional (and pan-regional, including the Midlands Engine) basis; but requires local ‘chapters’ of business leaders to lead in identifying opportunities for growth.

At a Combined Authority level, the Channel 4 consultation provides a template for how the three LEPs can work together to articulate a

Annex 1. Change in number of creative enterprises in GBSLEP local authorities, 2010-15

We now analyse the distribution of these creative enterprises across the individual local authorities in the GBSLEP area. Details of the number of creative enterprises in each local authority area can be found in the Annexes at the end of this report.

Birmingham

Over the five years to 2015, the most rapid increases in Creative Industries enterprises occurred in IT, Software & Computer Services; Architecture; and Publishing.

Overall, IT, Software & Computer Services provides more enterprises than other Creative Industries sectors: over 52% of the total (as defined by DCMS) in 2015.

Only Crafts and Music, Visual & Performing Arts saw falls in the number of enterprises. The fall in Crafts is perhaps indicative of some consolidation in the jewellery industry – and the Jewellery Quarter remains a nationally important location for this sector.

Bromsgrove

The number of sectors in which the number of creative enterprises did not change between 2010 and 2015 in Bromsgrove is perhaps indicative of both stability and low dynamism.

Architecture and IT, Software & Computer Services demonstrated the strongest rates of growth over the period.

Cannock Chase

The number of enterprises in Music, Visual & Performing Arts in Cannock Chase doubled between 2010 and 2015.

Design and IT, Software & Computer Services also saw strong performance over the same period, reflecting the strength of these sectors across the West Midlands.

There are, however, a number of Creative Industries sectors with no presence in Cannock Chase. Clearly, having some enterprises is a precondition of the locality achieving growth and jobs via these sectors.

East Staffordshire

From low bases, the number of enterprises in Architecture and Publishing in East Staffordshire doubled between 2010 and 2015.

Strong growth was also recorded in Design and IT, Software & Computer Services, while an overall increase of 24% in the number of creative enterprises is robust.

Lichfield

We observed some symmetry between the experiences East Staffordshire and Lichfield, which borders it to the south. Both geographies saw big increases, from low bases, in the number of enterprises in Architecture and Publishing between 2010 and 2015. Perhaps this suggests something specific to the Lichfield/East Staffordshire geography occurring in respect of these sectors.

The increases in Design and IT, Software & Computer Services enterprises in Lichfield between 2010 and 2015 are more typical of trends that we see across the West Midlands.

Redditch

Redditch shows growth in those sectors in which there is strength across the West Midlands, as demonstrated by increases in the number of enterprises between 2010 and 2015: Advertising & Marketing; Design; IT, Software & Computer Services.

Growth in these sectors offset falls in others, producing an overall increase of 5% in the number of enterprises within Creative Industries.

	Birmingham			Bromsgrove			Cannock Chase			East Staffordshire		
	2010	2015	% change	2010	2015	% change	2010	2015	% change	2010	2015	% change
Advertising & Marketing	215	240	12	35	35	0	15	15	0	25	25	0
Architecture	120	170	42	20	30	50	10	10	0	10	20	100
Crafts	150	125	-17	5	5	0	0	0		0	0	
Design: Product, Graphic and Fashion Design	150	185	23	40	40	0	20	30	50	15	25	67
Film, TV, Video, Radio & Photography	110	120	9	20	20	0	0	0		10	10	0
IT, Software, and Computer Services	830	1,315	58	160	215	34	65	85	31	120	160	33
Publishing	80	100	25	5	5	0	5	5	0	5	10	100
Museums, Galleries & Libraries	5	5	0	0	0		0	0		0	0	
Music, Visual & Performing Arts	240	230	-4	30	15	-50	5	10	100	25	10	-60
DCMS total	1,900	2,490	31	315	365	16	120	155	29	210	260	24
Tech Nation sectors	1,295	1,780	37	255	295	16	95	125	32	180	220	22

	Lichfield			Redditch			Solihull			Tamworth			Wyre Forest		
	2010	2015	% change	2010	2015	% change	2010	2015	% change	2010	2015	% change	2010	2015	% change
A&M	45	45	0	15	20	33	65	80	23	5	10	100	30	20	-33
Arch	15	25	67	10	5	-50	15	30	100	15	15	0	10	15	50
Crafts	5	5	0	5	0	-100	0	0		0	0		0	0	
Design	30	40	33	15	25	67	45	60	33	10	15	50	15	20	33
Film etc	15	5	-67	10	5	-50	20	15	-25	5	0	-100	15	15	0
IT	135	155	15	115	125	9	380	540	42	60	70	17	85	110	29
Publ.	5	15	200	5	5	0	15	10	-33	5	5	0	5	0	-100
MGL	0	0	-	0	0		0	0		0	0		0	0	
Music	30	30	0	10	10	0	35	50	43	10	5	-50	20	20	0
DCMS	280	320	14	185	195	5	575	785	37	110	120	9	180	200	11
Tech	205	230	12	160	165	3	520	690	33	105	115	10	125	140	12

Solihull

Solihull demonstrates strength in IT, Computer Services & Software, with 540 enterprises in this category in 2015 - a 42% increase on 2010, with IT comprising 69% of the total number of Creative Industries enterprises in the local authority.

Advertising & Marketing; Architecture; and Music, Visual & Performing Arts have also grown strongly.

Tamworth

While there are number of sectors in which the number of enterprises in Tamworth appears both low and static, the overall number of creative enterprises increased between 2010 and 2015.

This increase was driven by growth in Tamworth within sectors that demonstrate strength across much of the West Midlands: Advertising & Marketing, Design, and IT, Software & Computer Services.

Wyre Forest

Over half of Creative Industries enterprises in Wyre Forest operate in the IT, Software & Computer Services sector. This sector displayed strong growth between 2010 and 2015, as did Design: both are sectors in which there seems to be strength across the West Midlands.

The most rapid growth in Wyre Forest, however, occurred in Architecture. Therefore, while there are a number of sectors with a very low and static number of enterprises in Wyre Forest, there are pockets of vibrant creative industries activity in that district.

Change in Creative Industries employment across Greater Birmingham

We now review the change in number of Creative Industries workers across the different local authorities that compose GBSLEP between 2010 and 2015.

Because of ONS restrictions on reproducing employment data on the grounds of 'disclosure', we have not attempted to reproduce tables of employment by local authority – the rounding up (and down) to the nearest hundred involved would make the data confusing and, in some cases, meaningless. We have instead selected highlights from the data that illustrate notable trends in employment by Creative Industries sector or by geography.

Birmingham

The overall increase in Creative Industries employment in the city between 2010 and 2015, at 8% (1,100 jobs) is larger than the increase seen across the wider GBSLEP geography of 5 percent.

Design employment grew by 300% between 2010 and 2015, to 800 workers.

IT, Software & Computer Services remains, though, much the largest source of employment - growing 21% to 6,000 workers, representing two in every five Creative Industries jobs in the city and 22% of all Creative Industries employment in Greater Birmingham.

Advertising & Marketing, Architecture and Crafts also saw strong employment growth.

Falls were witnessed in Film, TV, Video, Radio & Photography; Publishing; Museums, Galleries & Libraries; and Music, Visual & Performing Arts.

Bromsgrove

Like Birmingham, Design employment in Bromsgrove grew strongly between 2010 and 2015 - a 227% increase to over 100 workers.

Also like Birmingham, IT, Software & Computer Services is much the largest source of employment - nearly 60% of total Creative Industries jobs in the borough.

While Crafts increased in Birmingham, it fell in Bromsgrove. But the other trends are further mirrors to Birmingham: rises in Advertising & Marketing and Architecture; falls in Film, TV, Video, Radio & Photography; Publishing; Museums, Galleries & Libraries; and Music, Visual & Performing Arts.

Cannock Chase

Cannock Chase is unusual in experiencing a fall in employment within the Creative Industries between 2010 and 2015.

Increases in employment in Advertising & Marketing; Architecture; and IT, Software & Computer Services were insufficient to compensate for falls in other sectors.

Film, TV, Video, Radio & Photography experienced the most dramatic fall in employment. In 2010, it was the second largest Creative Industries sector, with over 200 workers, while by 2015, it had contracted by 89%.

East Staffordshire

Dramatic increases in employment in Architecture and Design helped drive an overall increase of 19% in the DCMS Creative Industries sectors between 2010 and 2015, to 900 workers.

While IT, Software & Computer Services is the largest of the DCMS sectors, there was a dramatic fall in employment within the related, but somewhat broader, basket of Tech Nation sectors. The Digital Industries in East Staffordshire appear, therefore, to have contracted.

Lichfield

Again reflecting trends observed in East Staffordshire, Lichfield has seen a 3% fall in employment within IT, Software & Computer Services, and a 9% fall in the Digital Industries sector as defined by Tech Nation.

Overall, Creative Industries employment increased by 5% between 2010 and 2015. This has been assisted by strong growth in Advertising & Marketing; Architecture; and Design. The pattern of increases in these sectors, alongside falls in others, is consistent with that witnessed elsewhere in Greater Birmingham.

Redditch

Redditch is unusual in Greater Birmingham in seeing an increase in employment within Film, TV, Video, Radio & Photography - an 87% increase, to 100 workers.

The most spectacular increase, however, occurred in Design - a 540% rise in employment, to 100 workers.

Solihull

While IT, Software & Computer Services is much the largest source of employment within the creative industries in Solihull, it declined slightly by 1%, to 4,100 workers.

Solihull is unusual in witnessing a large increase in Music, Visual & Performing Arts, consistent with growth in demand from new production capacity and events held at NEC. The other trends were in line with the rest of Greater Birmingham and Solihull, with strong increases in Advertising & Marketing, Architecture, and Design.

Tamworth

Tamworth witnessed a decline in Creative Industries jobs between 2010 and 2015, due in large part to a fall in IT, Software & Computer Services,

in contrast to the rises in this sector witnessed in most other parts of Greater Birmingham.

Bright spots for Tamworth were Advertising & Marketing; Architecture; Design; and Publishing. Most of these sectors performed strongly across the GBSLEP area; but the increase in publishing is not widely seen across the West Midlands, which might be associated with a particular strength in Tamworth.

Wyre Forest

The fall in Creative Industries employment in Wyre Forest has been significant - a 22% fall between 2010 and 2015, to 600 workers.

While some sectors witnessed increases - in particular, Design rose by 110% - these failed to compensate for falls in other parts of the Creative Industries, with a major fall in Film, TV, Video, Radio & Photography being the biggest contributor to this overall decline.

Annex 2 - Birmingham

Figure 39. **Change in numbers of Creative Industries enterprises in Birmingham between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	215	240	12
Architecture	120	170	42
Crafts	150	125	-17
Design: Product, Graphic and Fashion Design	150	185	23
Film, TV, Video, Radio & Photography	110	120	9
IT, Software & Computer Services	830	1,315	58
Publishing	80	100	25
Museums, Galleries & Libraries	5	5	0
Music, Visual & Performing Arts	240	230	-4
DCMS Total	1,900	2,490	31
Tech Nation sectors	1,295	1,780	37

Source: NOMIS, ONS, BOP Analysis

Figure 40. **Change in Creative Industries employment in Birmingham between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	1,500	1,900	27
Architecture	800	1,100	29
Crafts	700	900	15
Design: Product, Graphic and Fashion Design	200	800	300
Film, TV, Video, Radio & Photography	2,300	1,500	-33
IT, Software & Computer Services	5,000	6,000	21
Publishing	800	500	-30
Museums, Galleries & Libraries	900	800	-11
Music, Visual & Performing Arts	1,700	1,500	-13
DCMS Total	14,000	15,100	8
Tech Nation sectors	10,100	10,700	6

Source: ONS/NOMIS/BOP Analysis

Annex 3 - Bromsgrove

Figure 41. **Change in numbers of Creative Industries enterprises in Bromsgrove between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	35	35	0
Architecture	20	30	50
Crafts	5	5	0
Design: Product, Graphic and Fashion Design	40	40	0
Film, TV, Video, Radio & Photography	20	20	0
IT, Software & Computer Services	160	215	34
Publishing	5	5	0
Museums, Galleries & Libraries	0	0	
Music, Visual & Performing Arts	30	15	-50
DCMS Total	315	365	16
Tech Nation sectors	255	295	16

Source: NOMIS, ONS, BOP Analysis

Figure 42. **Change in Creative Industries employment in Bromsgrove between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	100	100	121
Architecture	0	0	165
Crafts	0	0	-97
Design: Product, Graphic and Fashion Design	0	100	227
Film, TV, Video, Radio & Photography	100	100	-38
IT, Software & Computer Services	600	900	51
Publishing	200	100	-32
Museums, Galleries & Libraries	100	100	-11
Music, Visual & Performing Arts	100	0	-49
DCMS Total	1,200	1,600	25
Tech Nation sectors	900	1,200	26

Source: ONS/NOMIS/BOP Analysis

Annex 4 –Cannock Chase

Figure 43. **Change in the number of Creative Industries enterprises in Cannock Chase between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	15	15	0
Architecture	10	10	0
Crafts	0	0	0
Design: Product, Graphic and Fashion Design	20	30	50
Film, TV, Video, Radio & Photography	0	0	0
IT, Software & Computer Services	65	85	31
Publishing	5	5	0
Museums, Galleries & Libraries	0	0	0
Music, Visual & Performing Arts	5	10	100
DCMS Total	120	155	29
Tech Nation sectors	95	125	32

Source: NOMIS, ONS, BOP Analysis

Figure 44. **Change in Creative Industries employment in Cannock Chase between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	0	0	86
Architecture	0	100	71
Crafts	0	0	0
Design: Product, Graphic and Fashion Design	100	100	-1
Film, TV, Video, Radio & Photography	200	0	-89
IT, Software & Computer Services	500	600	14
Publishing	0	0	-75
Museums, Galleries & Libraries	100	0	-19
Music, Visual & Performing Arts	0	0	0
DCMS Total	1,000	900	-14
Tech Nation sectors	700	700	3

Source: ONS/NOMIS/BOP Analysis

Annex 5 –East Staffordshire

Figure 45. **Change in the number of Creative Industries enterprises in East Staffordshire between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	25	25	0
Architecture	10	20	100
Crafts	0	0	0
Design: Product, Graphic and Fashion Design	15	25	67
Film, TV, Video, Radio & Photography	10	10	0
IT, Software & Computer Services	120	160	33
Publishing	5	10	100
Museums, Galleries & Libraries	0	0	
Music, Visual & Performing Arts	25	10	-60
DCMS Total	210	260	24
Tech Nation sectors	180	220	22

Source: NOMIS, ONS, BOP Analysis

Figure 46. **Change in Creative Industries employment in East Staffordshire between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	0	0	44
Architecture	0	100	312
Crafts	0	0	0
Design: Product, Graphic and Fashion Design	0	100	613
Film, TV, Video, Radio & Photography	100	100	5
IT, Software & Computer Services	500	500	3
Publishing	0	0	9
Museums, Galleries & Libraries	100	100	-9
Music, Visual & Performing Arts	100	0	-38
DCMS Total	800	900	19
Tech Nation sectors	2,200	700	-66

Source: ONS/NOMIS/BOP Analysis

Annex 6 - Lichfield

Figure 47. **Change in the number of Creative Industries enterprises in Lichfield between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	45	45	0
Architecture	15	25	67
Crafts	5	5	0
Design: Product, Graphic and Fashion Design	30	40	33
Film, TV, Video, Radio & Photography	15	5	-67
IT, Software & Computer Services	135	155	15
Publishing	5	15	200
Museums, Galleries & Libraries	0	0	
Music, Visual & Performing Arts	30	30	0
DCMS Total	280	320	14
Tech Nation sectors	205	230	12

Source: NOMIS, ONS, BOP Analysis

Figure 48. **Change in Creative Industries employment in Lichfield between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	0	100	176
Architecture	100	100	40
Crafts	0	0	0
Design: Product, Graphic and Fashion Design	0	100	423
Film, TV, Video, Radio & Photography	100	0	-49
IT, Software & Computer Services	800	800	-3
Publishing	0	0	-55
Museums, Galleries & Libraries	100	100	-13
Music, Visual & Performing Arts	100	100	2
DCMS Total	1,300	1,300	5
Tech Nation sectors	1,400	1,300	-9

Source: ONS/NOMIS/BOP Analysis

Annex 7 - Redditch

Figure 49. Change in the number of Creative Industries enterprises in Redditch between 2010 and 2015

	2010	2015	% change
Advertising & Marketing	15	20	33
Architecture	10	5	-50
Crafts	5	0	-100
Design: Product, Graphic and Fashion Design	15	25	67
Film, TV, Video, Radio & Photography	10	5	-50
IT, Software & Computer Services	115	125	9
Publishing	5	5	0
Museums, Galleries & Libraries	0	0	
Music, Visual & Performing Arts	10	10	0
DCMS Total	185	195	5
Tech Nation sectors	160	165	3

Source: NOMIS, ONS, BOP Analysis

Figure 50. Change in Creative Industries employment in Redditch between 2010 and 2015

	2010	2015	% change
Advertising & Marketing	0	0	41
Architecture	0	0	-14
Crafts	0	0	-100
Design: Product, Graphic and Fashion Design	0	100	540
Film, TV, Video, Radio & Photography	0	100	87
IT, Software & Computer Services	500	500	0
Publishing	100	0	-65
Museums, Galleries & Libraries	0	0	-45
Music, Visual & Performing Arts	0	0	2
DCMS Total	800	800	4
Tech Nation sectors	1,300	1,200	-7

Source: ONS/NOMIS/BOP Analysis

Annex 8 - Solihull

Figure 51. **Change in the number of Creative Industries enterprises in Solihull between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	65	80	23
Architecture	15	30	100
Crafts	0	0	
Design: Product, Graphic and Fashion Design	45	60	33
Film, TV, Video, Radio & Photography	20	15	-25
IT, Software & Computer Services	380	540	42
Publishing	15	10	-33
Museums, Galleries & Libraries	0	0	
Music, Visual & Performing Arts	35	50	43
DCMS Total	575	785	37
Tech Nation sectors	520	690	33

Source: NOMIS, ONS, BOP Analysis

Figure 52. **Change in Creative Industries employment in Solihull between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	500	500	22
Architecture	100	100	56
Crafts	0	0	100
Design: Product, Graphic and Fashion Design	0	100	233
Film, TV, Video, Radio & Photography	500	200	-63
IT, Software & Computer Services	4,200	4,100	-1
Publishing	0	100	35
Museums, Galleries & Libraries	300	300	-13
Music, Visual & Performing Arts	100	300	288
DCMS Total	5,600	5,700	1
Tech Nation sectors	5,300	5,400	1

Source: ONS/NOMIS/BOP Analysis

Annex 9 - Tamworth

Figure 53. **Change in the number of Creative Industries enterprises in Tamworth between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	5	10	100
Architecture	15	15	0
Crafts	0	0	
Design: Product, Graphic and Fashion Design	10	15	50
Film, TV, Video, Radio & Photography	5	0	-100
IT, Software & Computer Services	60	70	17
Publishing	5	5	0
Museums, Galleries & Libraries	0	0	
Music, Visual & Performing Arts	10	5	-50
DCMS Total	110	120	9
Tech Nation sectors	105	115	10

Source: NOMIS, ONS, BOP Analysis

Figure 54. **Change in Creative Industries employment in Tamworth between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	0	100	42
Architecture	0	100	86
Crafts	0	0	
Design: Product, Graphic and Fashion Design	0	100	400
Film, TV, Video, Radio & Photography	100	100	-31
IT, Software & Computer Services	500	300	-27
Publishing	0	100	967
Museums, Galleries & Libraries	100	0	-47
Music, Visual & Performing Arts	0	0	-76
DCMS Total	700	700	-6
Tech Nation sectors	700	700	2

Source: ONS/NOMIS/BOP Analysis

Annex 10 - Wyre Forest

Figure 55. **Change in the number of Creative Industries enterprises in Wyre Forest between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	30	20	-33
Architecture	10	15	50
Crafts	0	0	
Design: Product, Graphic and Fashion Design	15	20	33
Film, TV, Video, Radio & Photography	15	15	0
IT, Software & Computer Services	85	110	29
Publishing	5	0	-100
Museums, Galleries & Libraries	0	0	
Music, Visual & Performing Arts	20	20	0
DCMS Total	180	200	11
Tech Nation sectors	125	140	12

Source: NOMIS, ONS, BOP Analysis

Figure 56. **Change in Creative Industries employment in Wyre Forest between 2010 and 2015**

	2010	2015	% change
Advertising & Marketing	0	0	84
Architecture	0	0	35
Crafts	0	0	
Design: Product, Graphic and Fashion Design	0	0	110
Film, TV, Video, Radio & Photography	200	0	-84
IT, Software & Computer Services	300	300	2
Publishing	0	0	-74
Museums, Galleries & Libraries	100	100	50
Music, Visual & Performing Arts	0	0	-36
DCMS Total	800	600	-22
Tech Nation sectors	500	500	7

Source: ONS/NOMIS/BOP Analysis

Annex 11 – DCMS Creative Industries SIC Codes

The DCMS Creative Industries SIC codes are:

Advertising & Marketing

- Public relations and communication activities
- Advertising agencies
- Media representation

Architecture

- Architectural activities

Crafts

- Manufacture of jewellery and related articles

Design: product, graphic and fashion design

- Specialised design activities

Film, TV, Video, Radio & Photography

- Motion picture, video and television programme production activities
- Motion picture, video and television programme postproduction
- Motion picture, video and television programme distribution
- Motion picture projection activities
- Radio broadcasting
- Television programming and broadcasting activities
- Photographic activities

IT, Software & Computer Services

- Publishing of computer games
- Other software publishing
- Computer programming activities

- Computer consultancy activities

Publishing

- Book publishing
- Publishing of directories and mailing lists
- Publishing of newspapers
- Publishing of journals and periodicals
- Other publishing activities
- Translation and interpretation activities

Museums, Galleries and Libraries

- Library and archive activities
- Museum activities

Music, Performing and Visual Arts

- Sound recording and music publishing activities
- Cultural education
- Performing arts
- Support activities to performing arts
- Artistic creation
- Operation of arts facilities

Annex 12 - DCMS Standard Occupation Codes (SOC)

Advertising & Marketing

- Marketing and Sales Directors
- Advertising and Public Relations Directors
- Public Relations Professionals
- Advertising Accounts Managers and Creative Directors
- Marketing Associate Professionals

Architecture

- Architects
- Town Planning
- Chartered Architectural Technologists
- Architectural and Town Planning Technicians

Crafts²⁴

- Smiths and Forge Workers
- Weavers and Knitters
- Glass and Ceramics Makers, Decorators and Finishers
- Furniture Makers and Other Craft Woodworkers
- Other Skilled Trades Not Elsewhere Classified

Design: Product, Graphic and Fashion Design

- Graphic Designers

- Product, Clothing and Related Designers

Film, TV, Video, Radio & Photography

- Arts Officers, Producers and Directors
- Photographers, Audio-Visual and Broadcasting Equipment Operators

IT, Software & Computer Services

- Information Technology and Telecommunications Directors
- IT Business Analysts, Architects and Systems Designers
- Programmers and Software Development Professionals
- Web Design and Development Professionals

Publishing

- Journalists, Newspaper and Periodical Editors
- Authors, Writers and Translators

Museums, Galleries & Libraries

- Librarians
- Archivists and Curators

Music, Visual & Performing Arts

- Artists
- Actors, Entertainers and Presenters
- Dancers and Choreographers
- Musicians

²⁴ One quirk of the DCMS groupings is that the crafts SIC grouping focuses on jewellery, while the crafts SOC grouping has a broader focus.

Annex 13 – Tech Nation SIC codes

Figure 57. The SIC codes comprising ‘Digital Industries’ as defined by Tech Nation.

SIC code	Description	Included in DCMS Creative Industries definition?
2620	Manufacture of computers and peripheral equipment	N
5821	Publishing of computer games	Y
5829	Other software publishing	Y
6110	Wired telecommunications activities	N
6120	Wireless telecommunications activities	N
6130	Satellite telecommunications activities	N
6190	Other telecommunications activities	N
6201	Computer programming activities	Y
6202	Computer consultancy activities	Y
6203	Computer facilities management activities	N
6209	Other IT & computer service activities	N
6311	Data processing, hosting & related activities	N
6312	Web portals	N
9511	Repair of computers & peripheral equipment	N